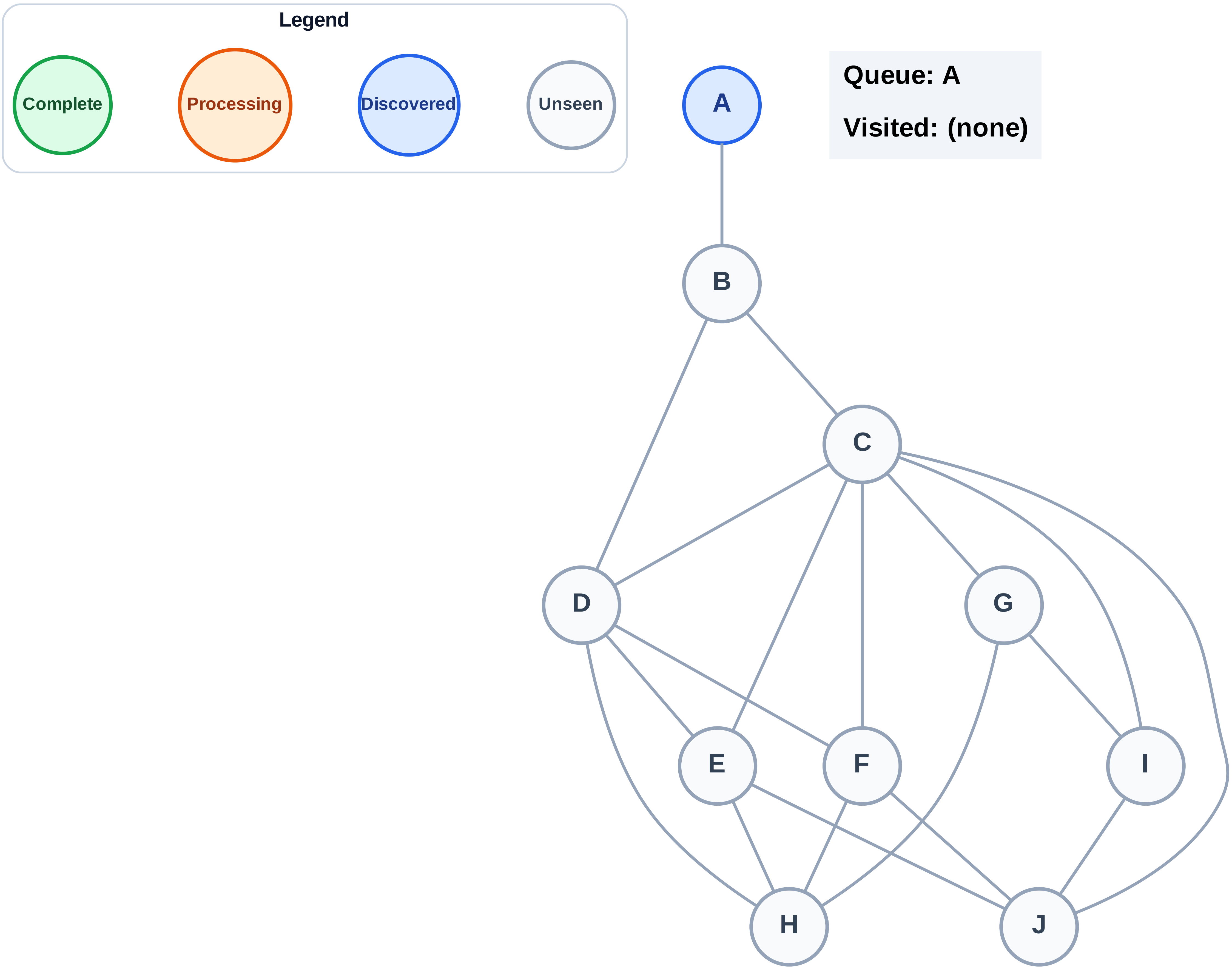
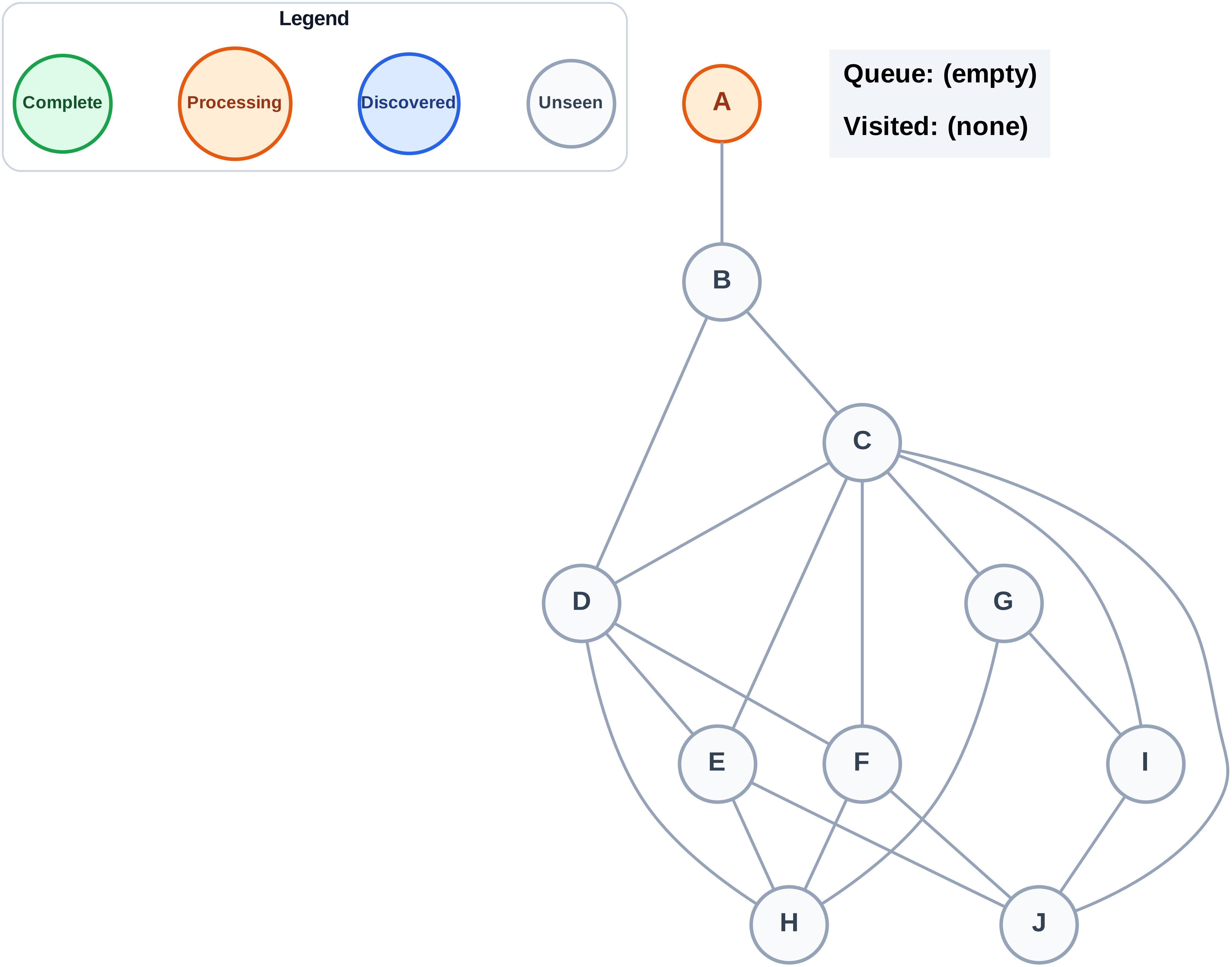
BFS Traversal

Step 1: Start at A



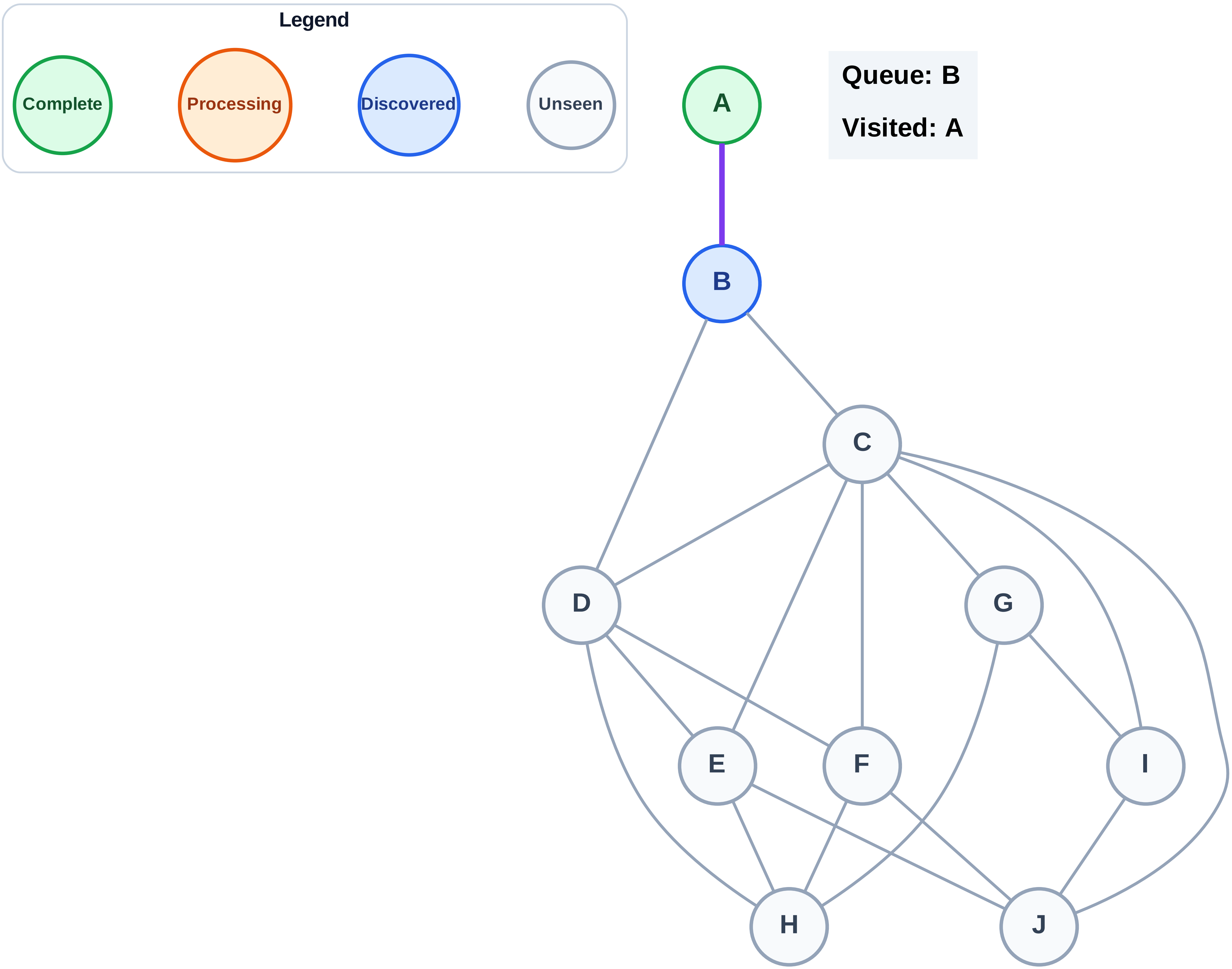
BFS Traversal

Step 2: Process node A



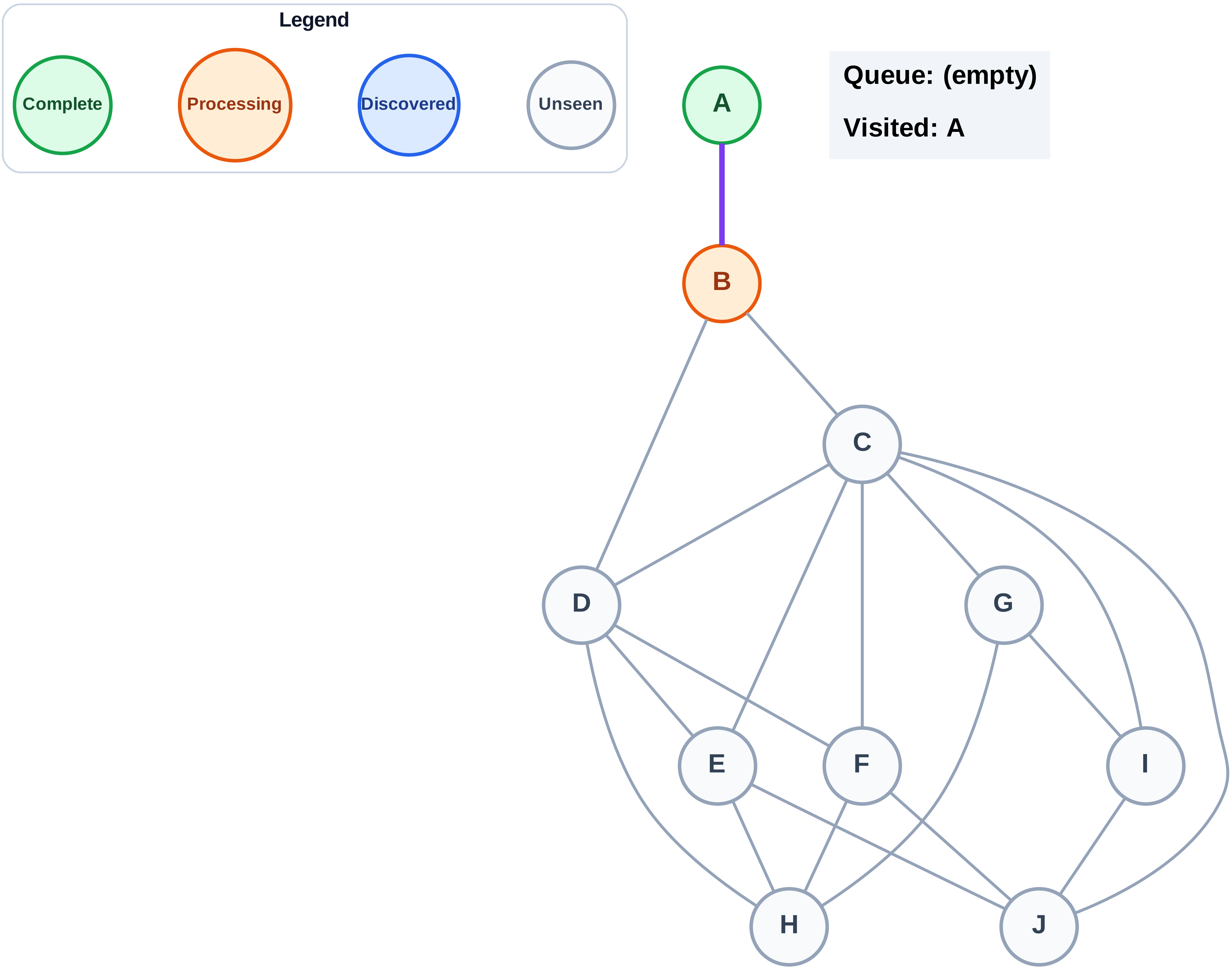
BFS Traversal

Step 3: Discover B from A



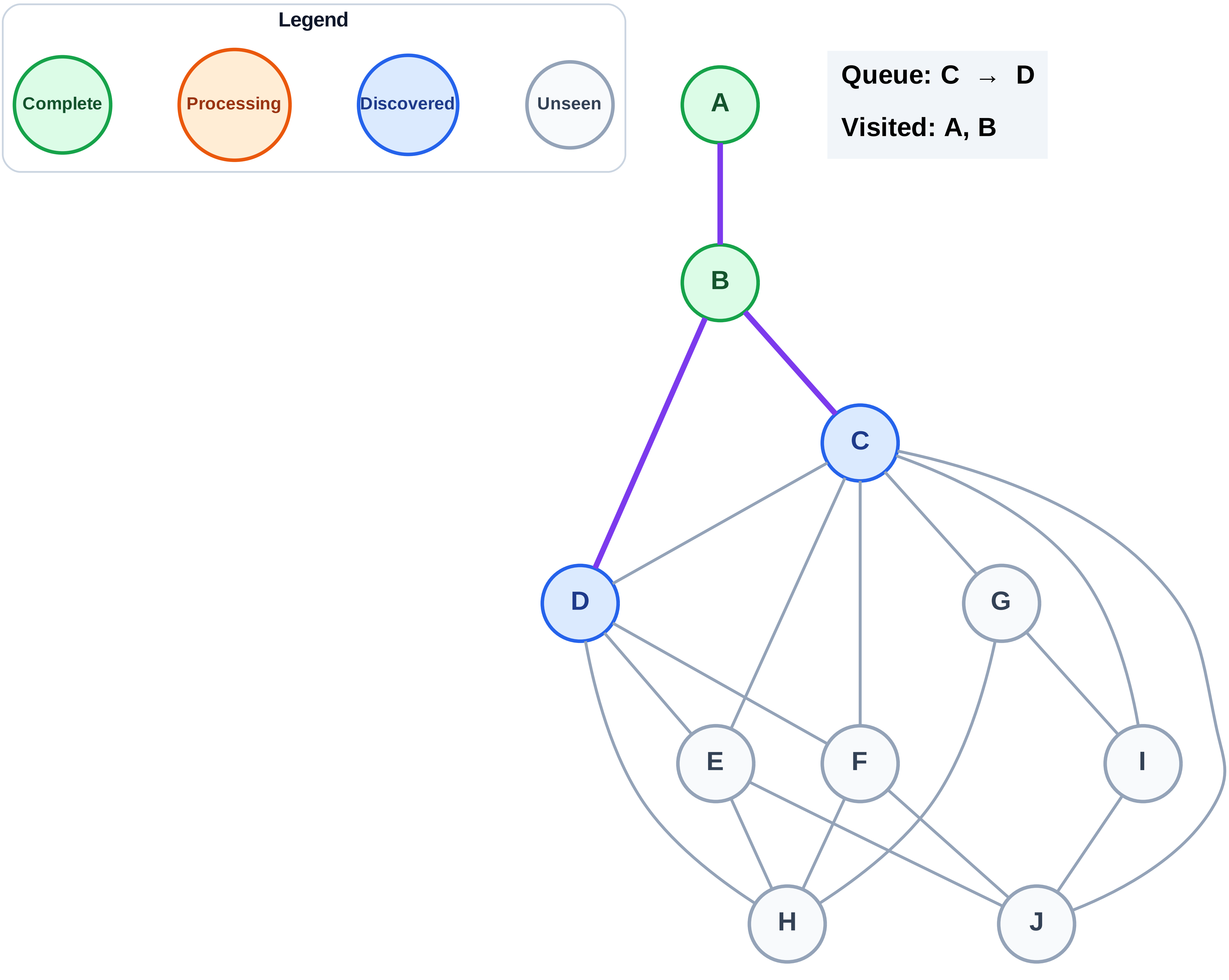
BFS Traversal

Step 4: Process node B



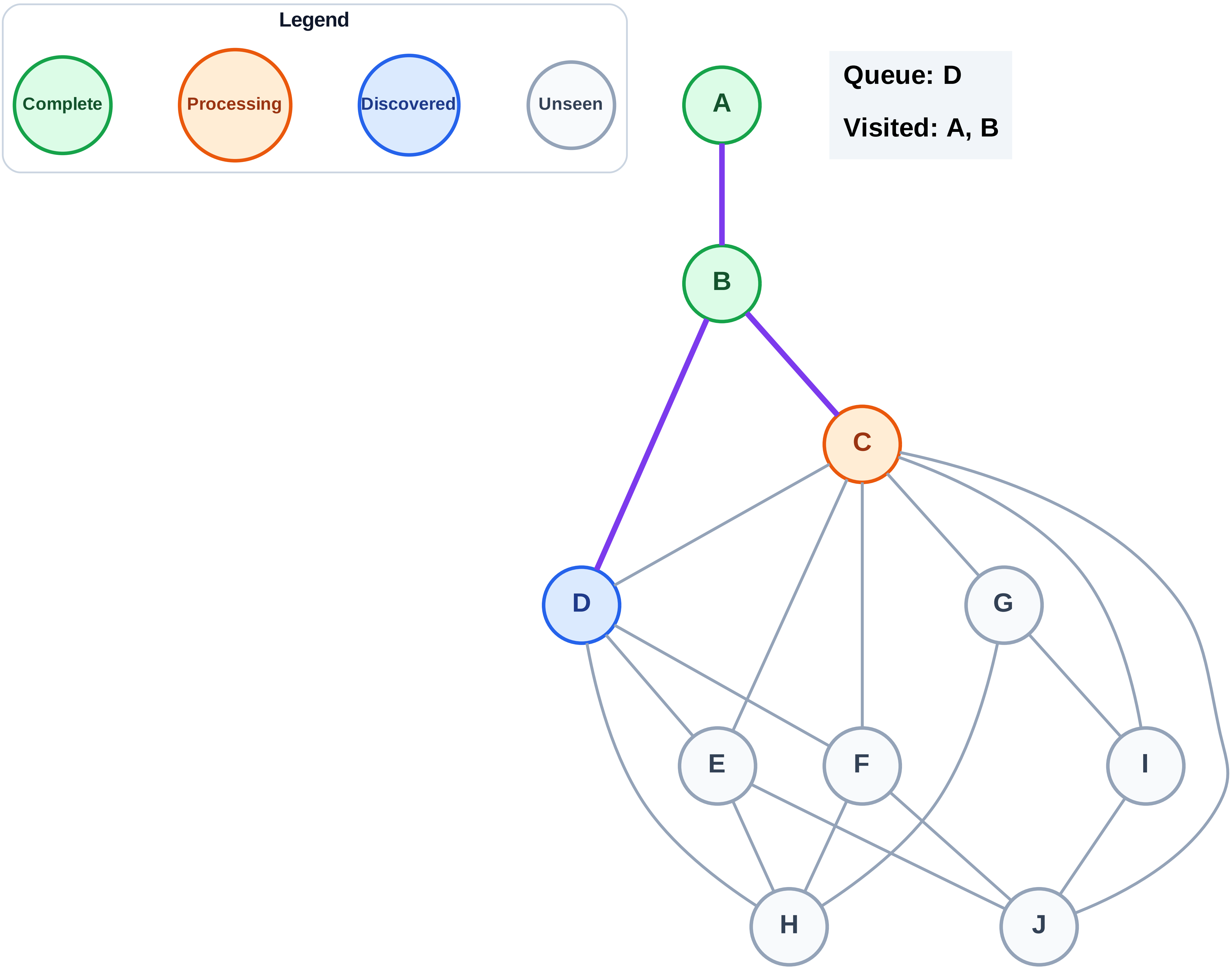
BFS Traversal

Step 5: Discover C, D from B



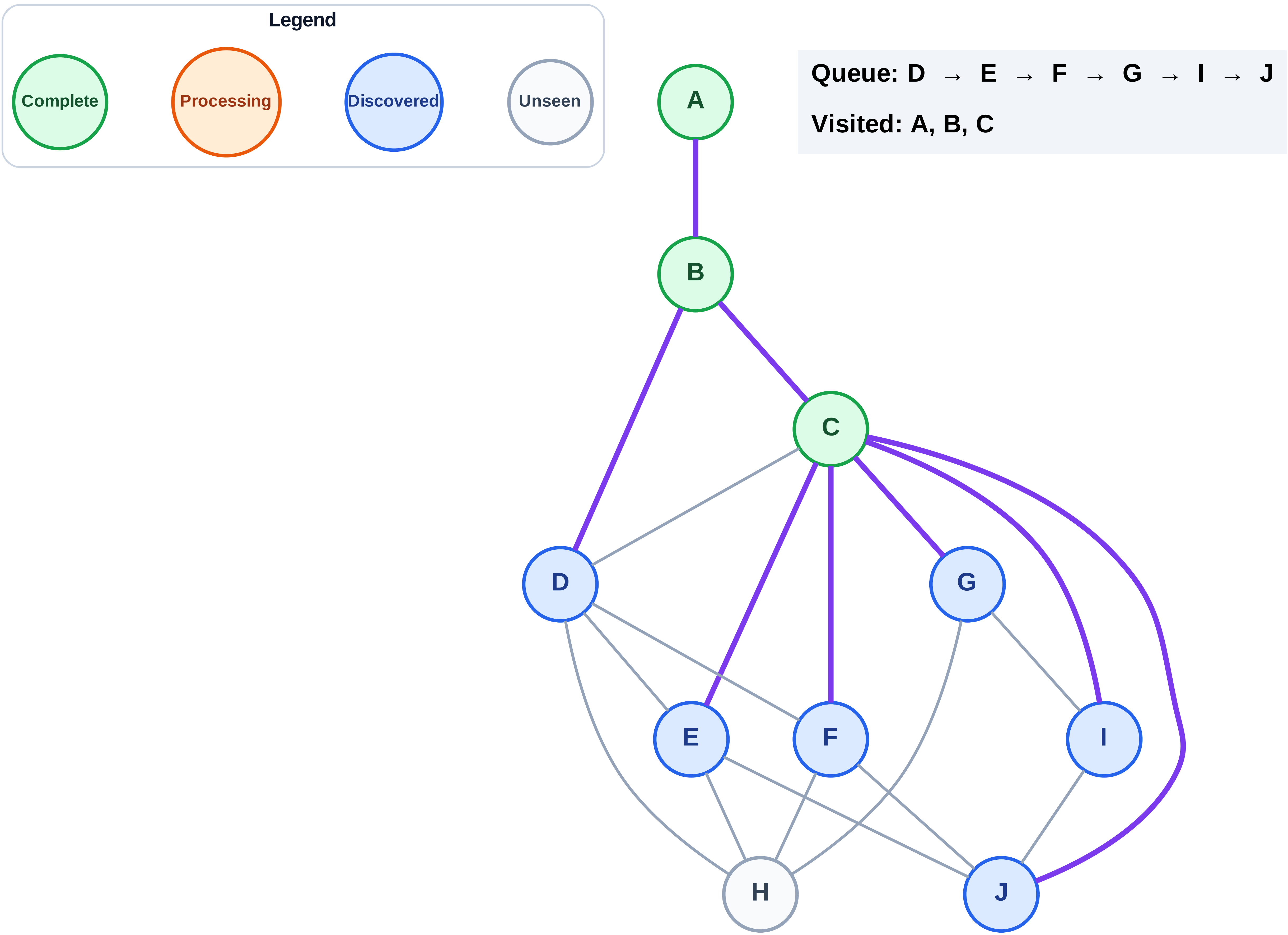
BFS Traversal

Step 6: Process node C



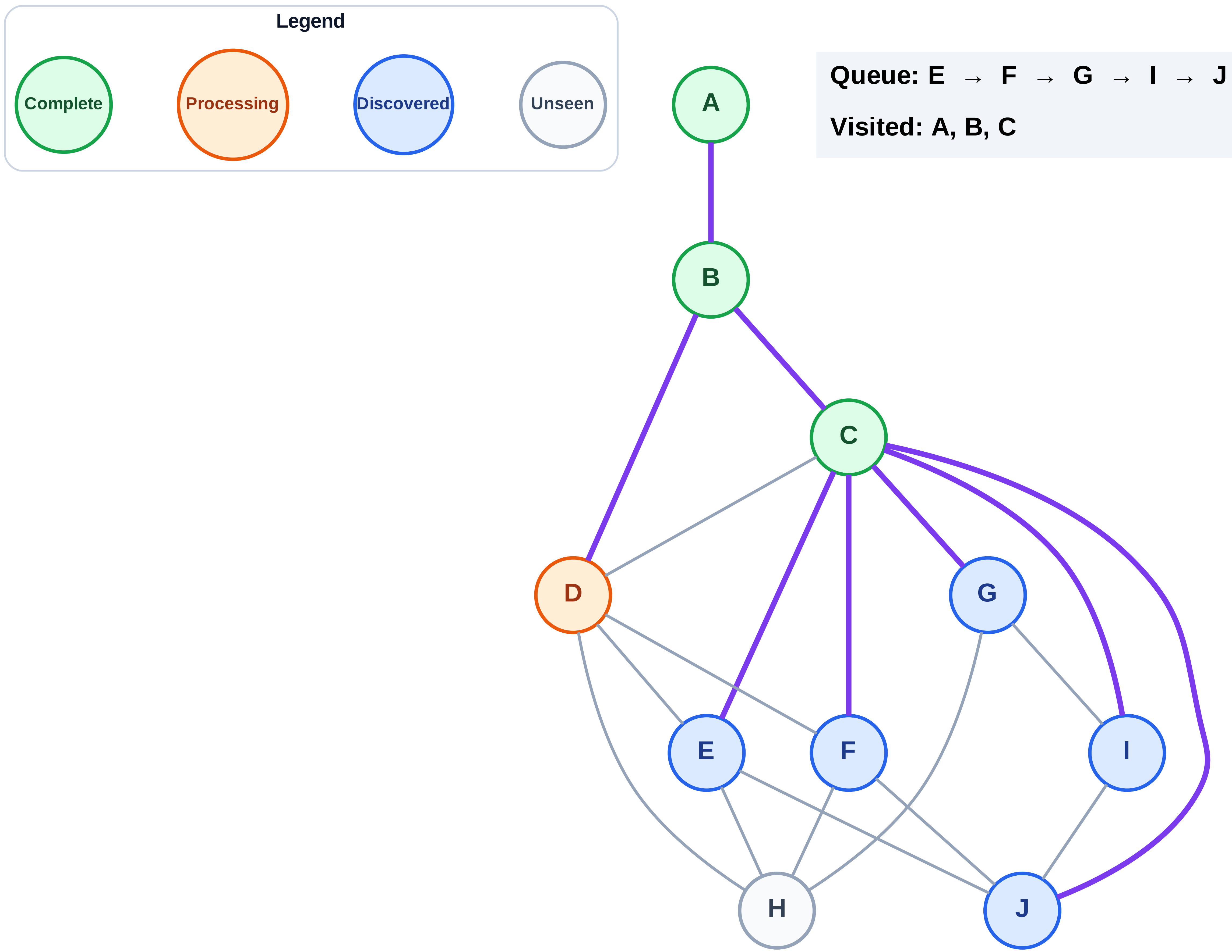
BFS Traversal

Step 7: Discover E, F, G, I, J from C



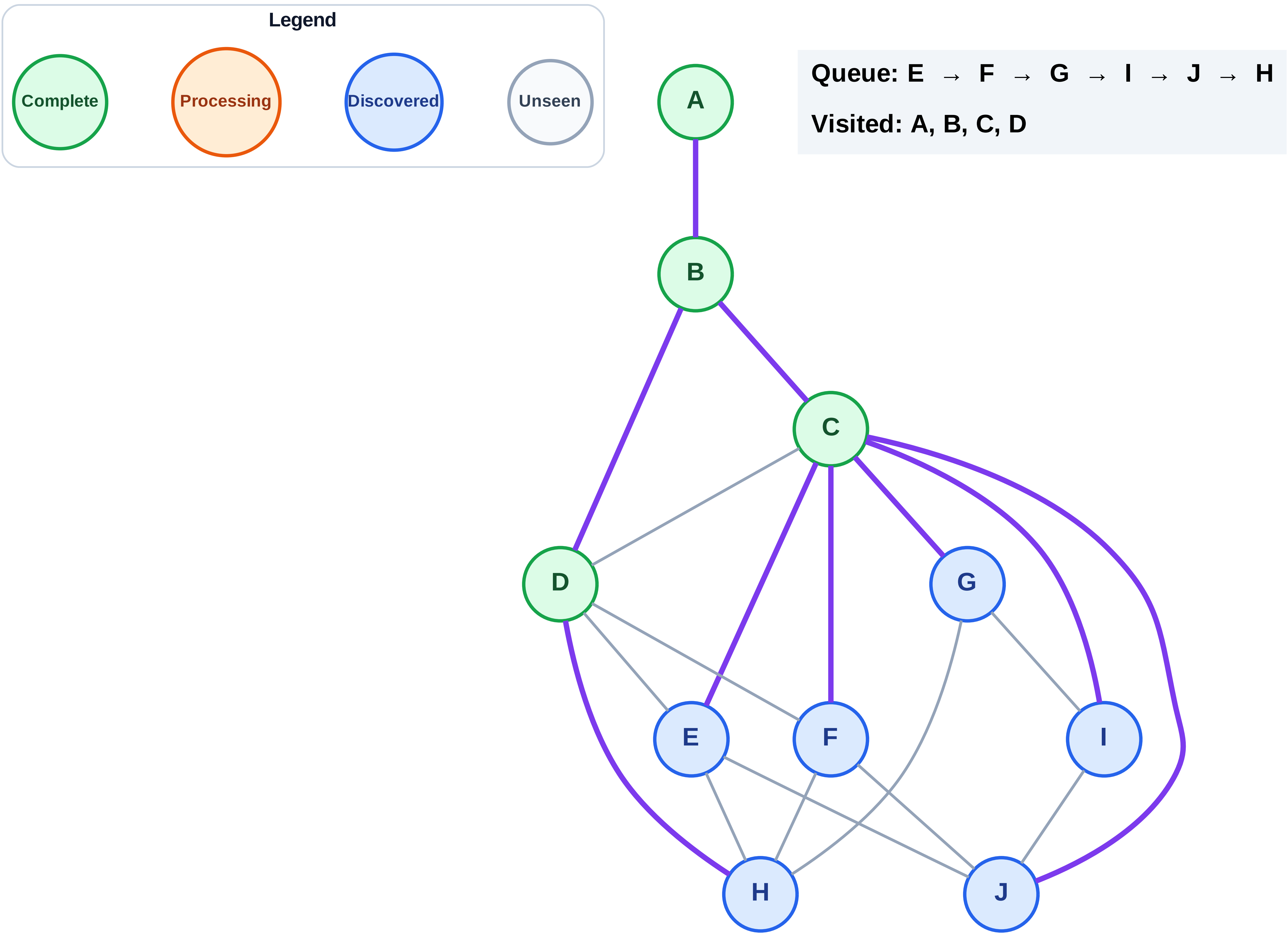
BFS Traversal

Step 8: Process node D



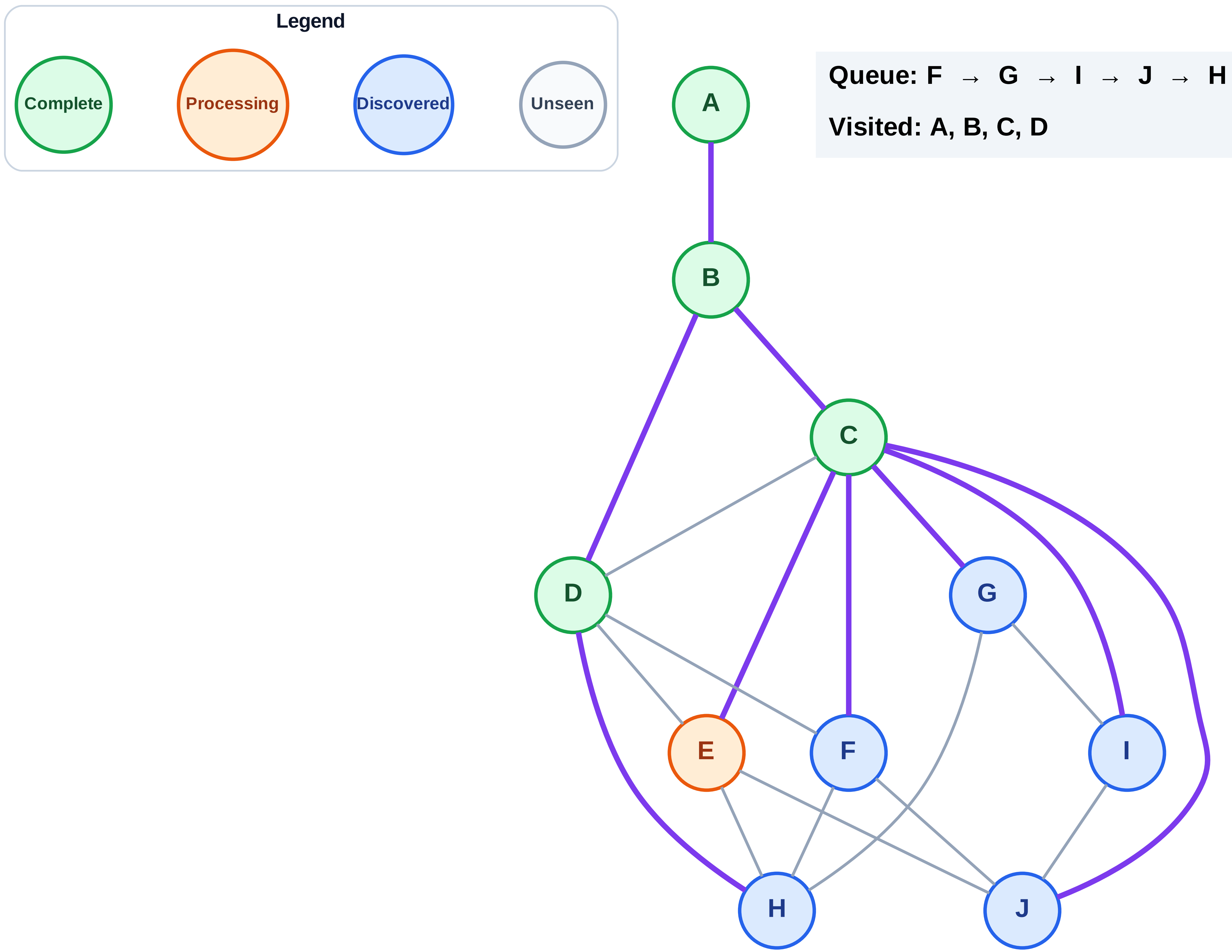
BFS Traversal

Step 9: Discover H from D



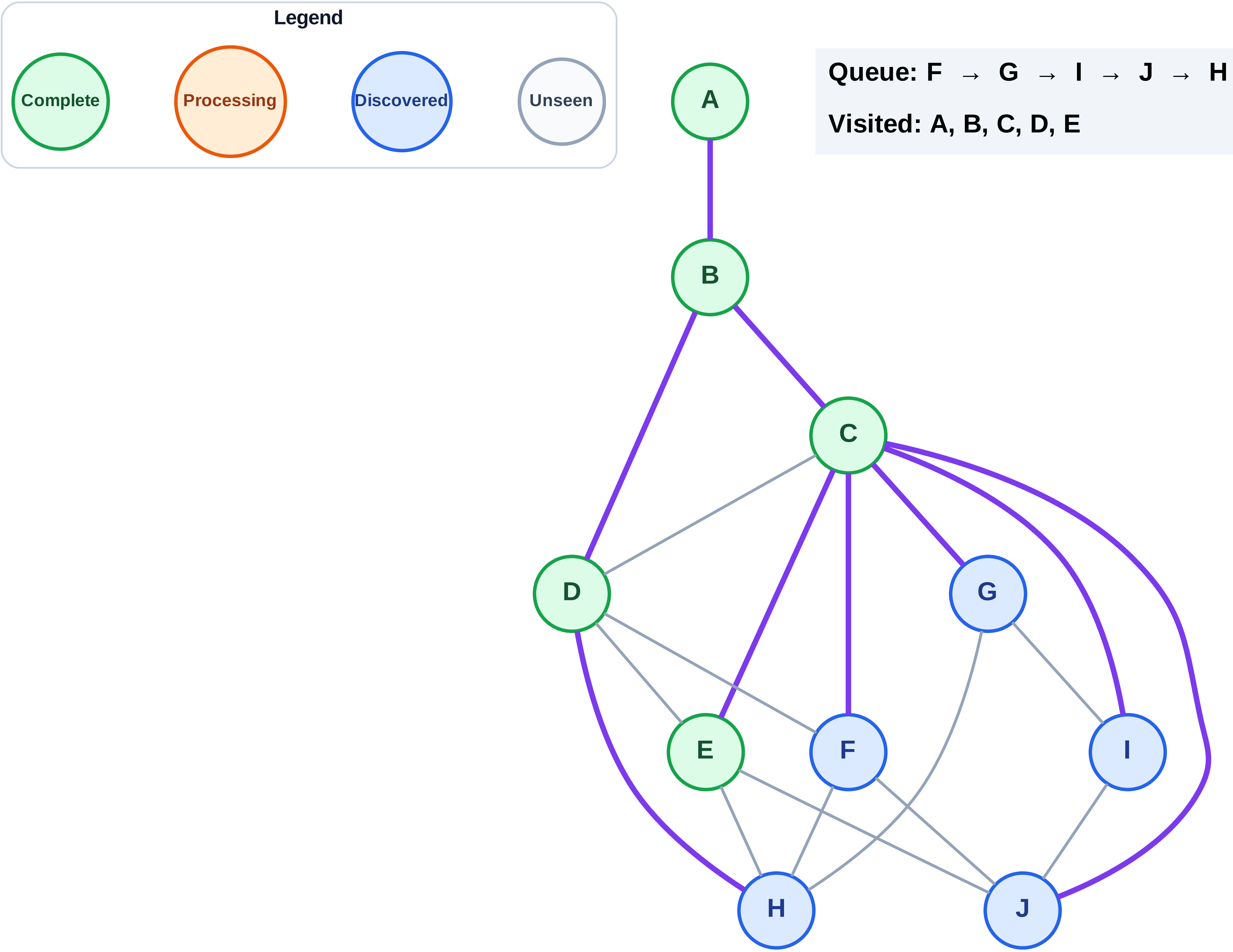
BFS Traversal

Step 10: Process node E



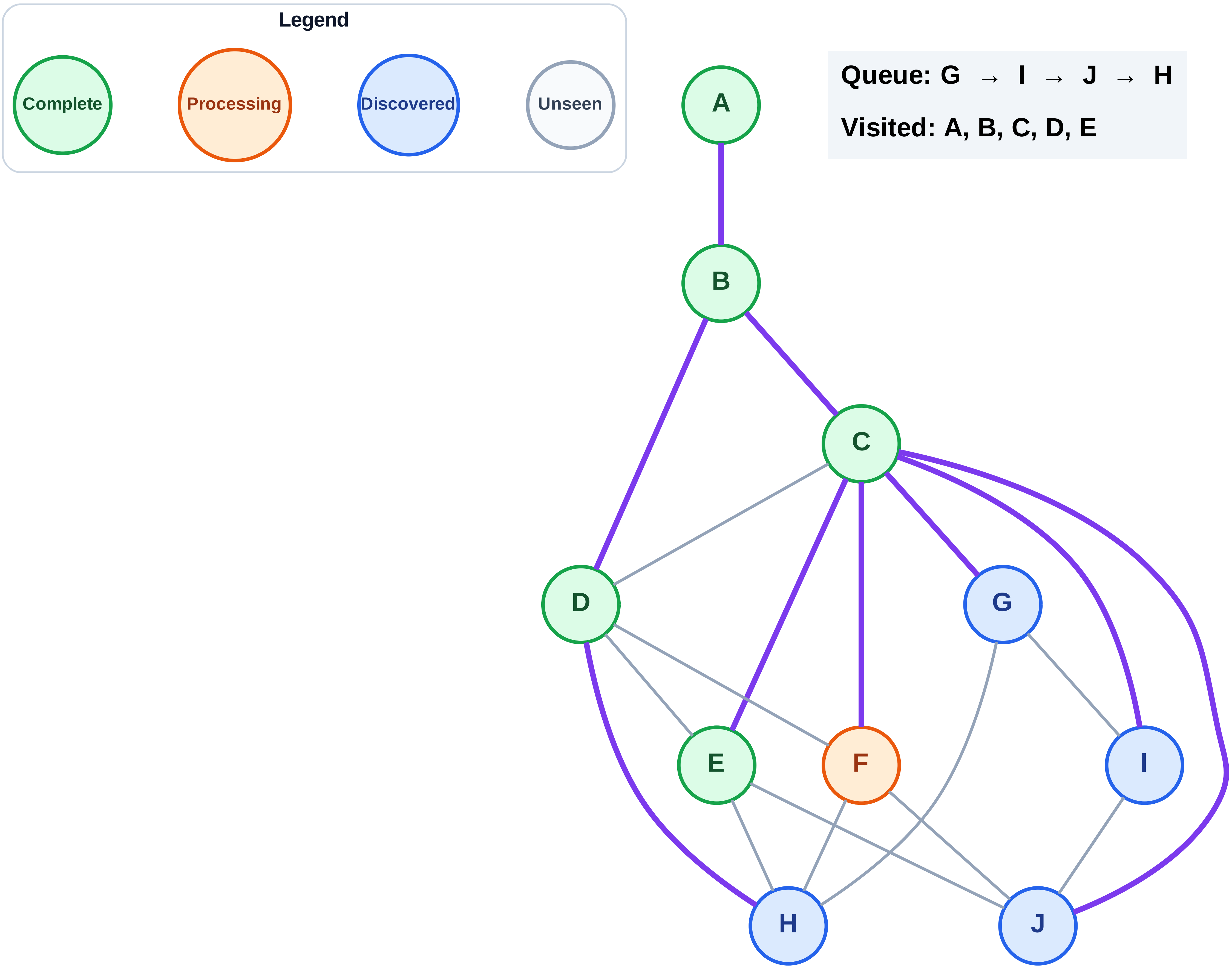
BFS Traversal

Step 11: No new neighbors from E



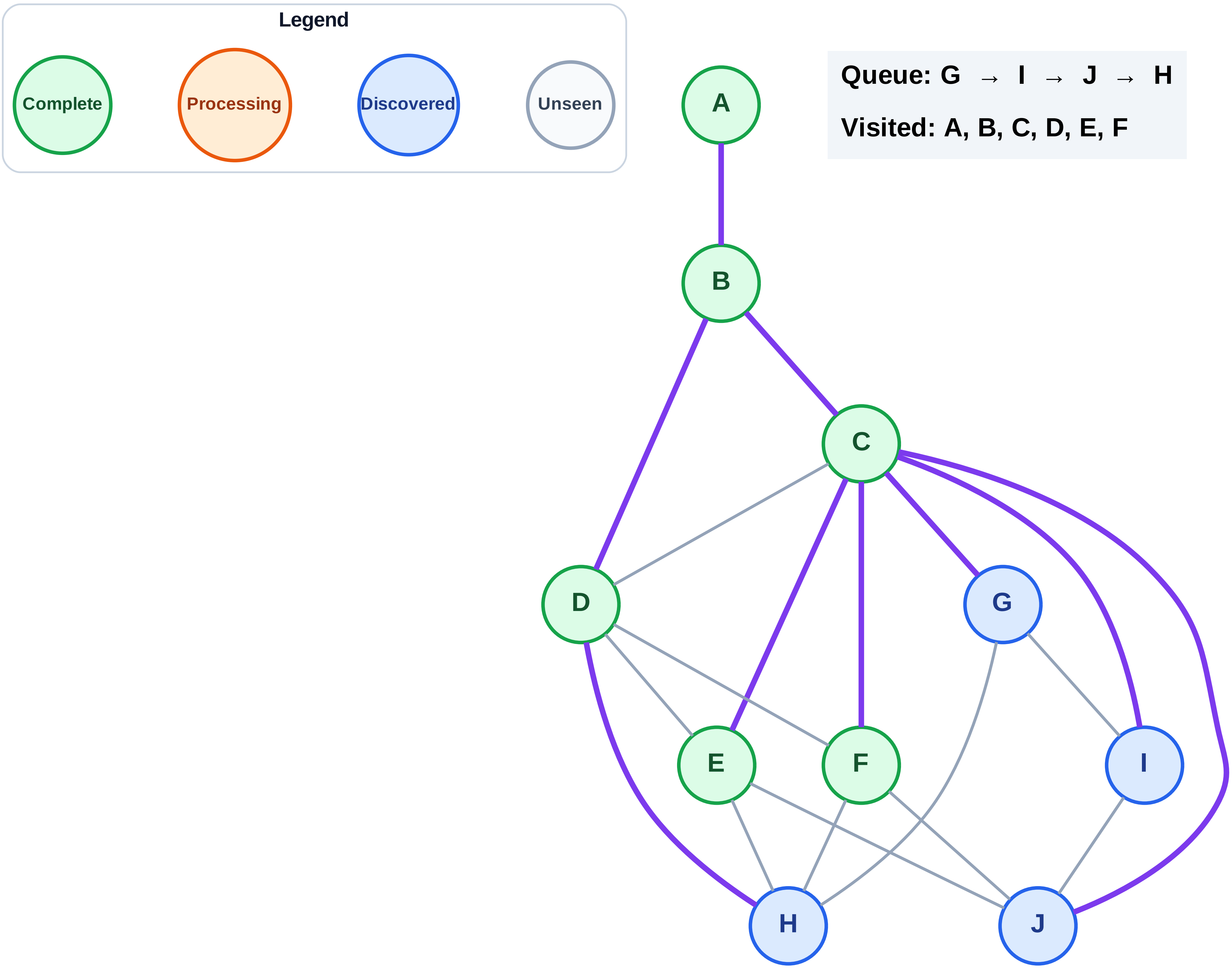
BFS Traversal

Step 12: Process node F



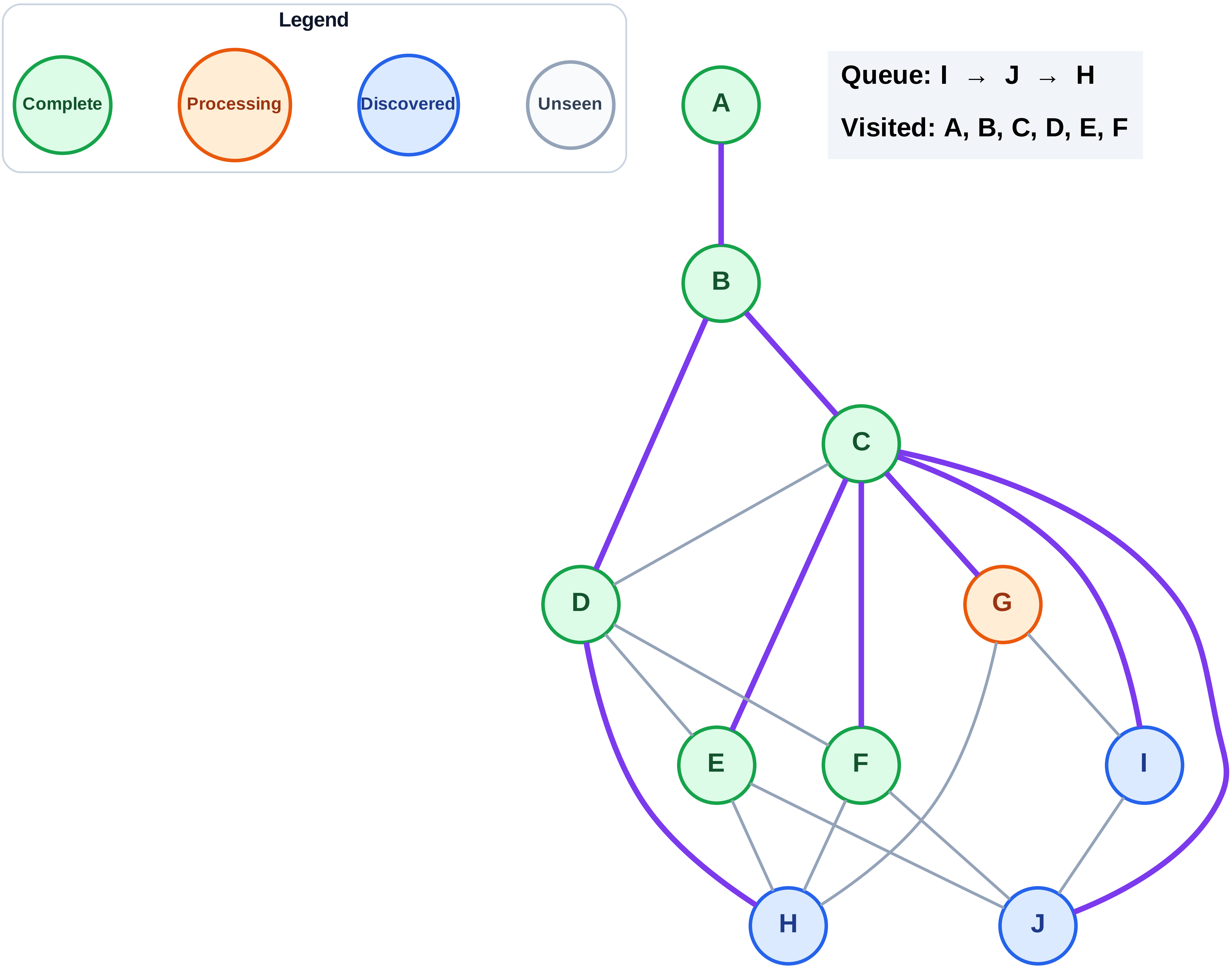
BFS Traversal

Step 13: No new neighbors from F



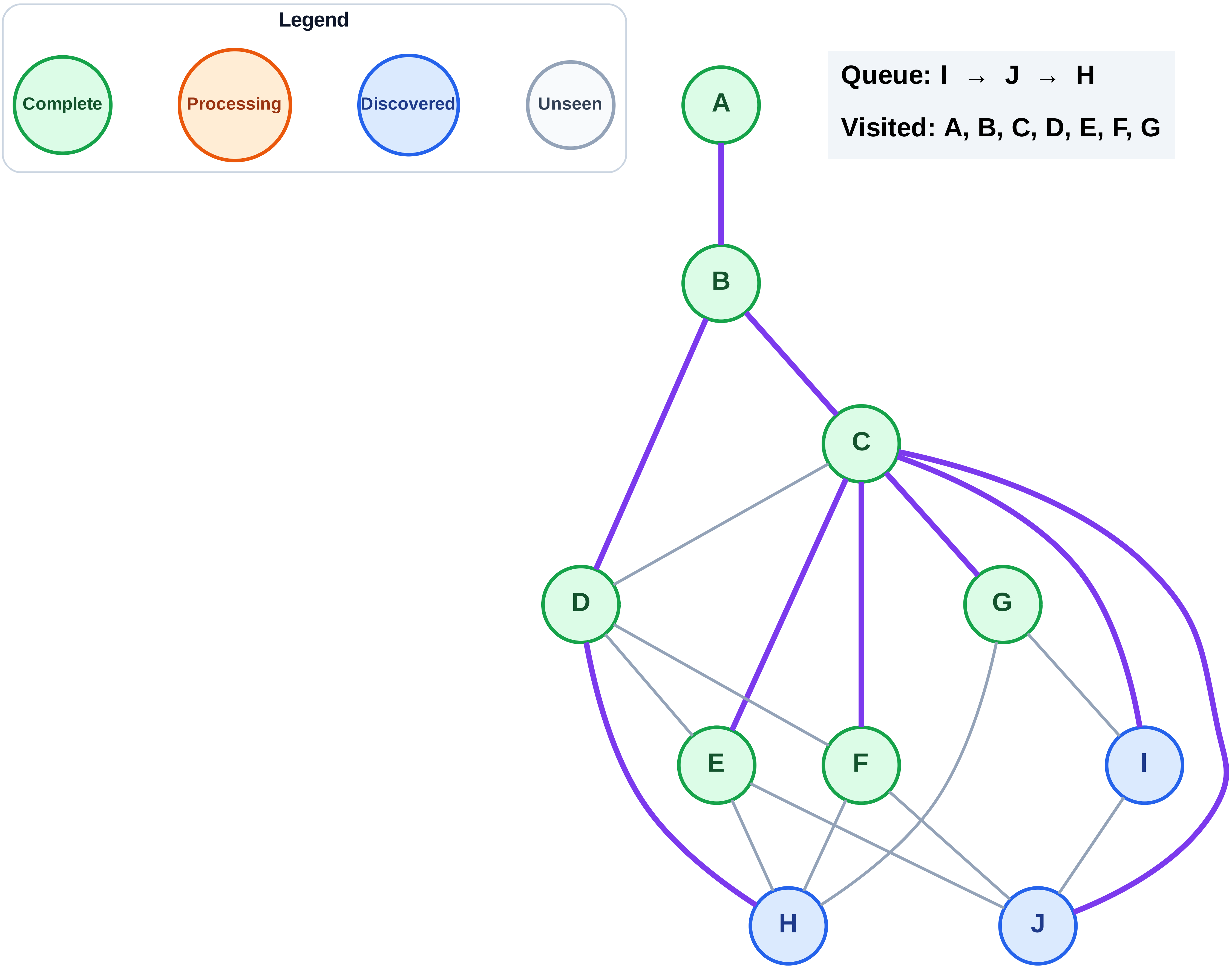
BFS Traversal

Step 14: Process node G



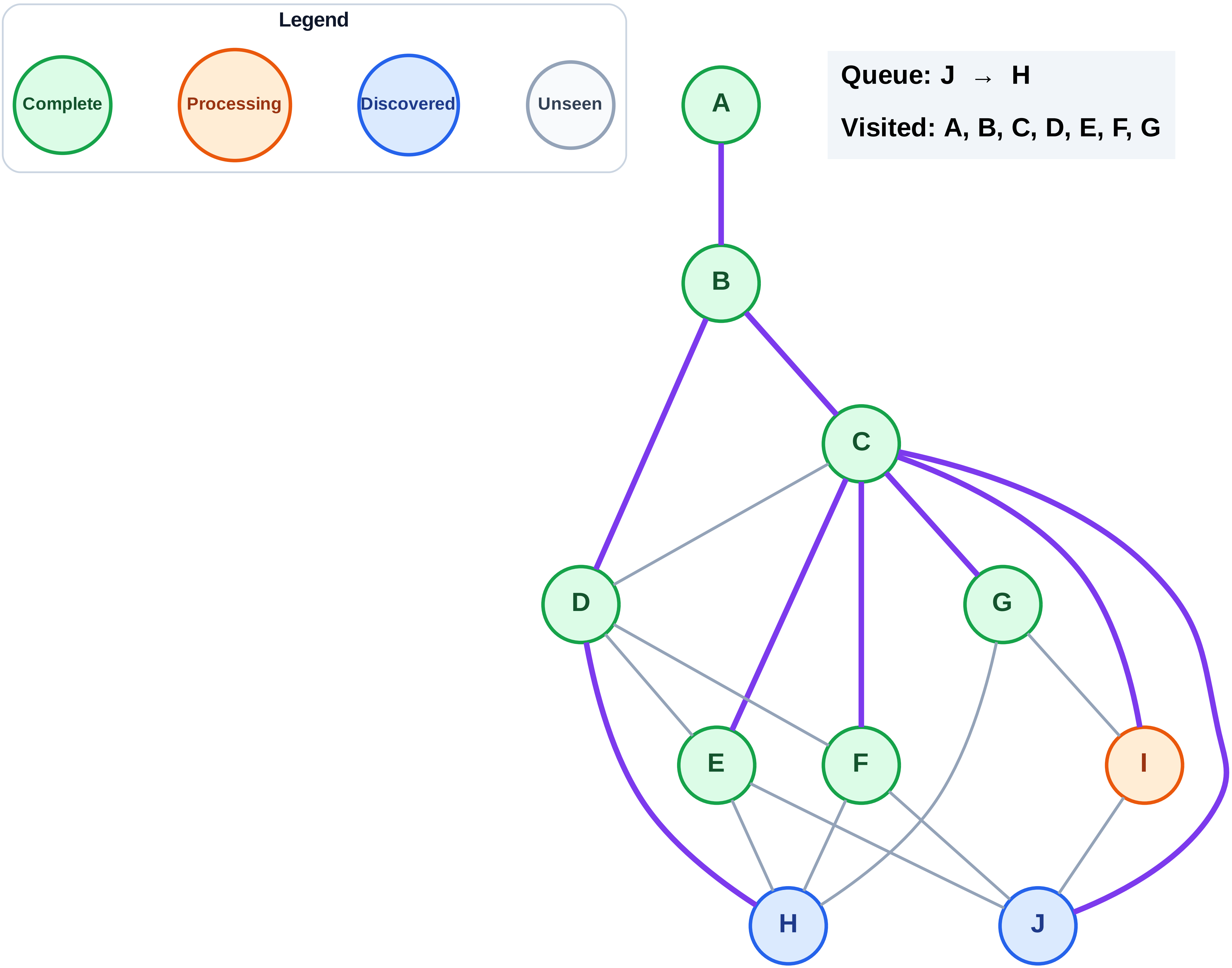
BFS Traversal

Step 15: No new neighbors from G



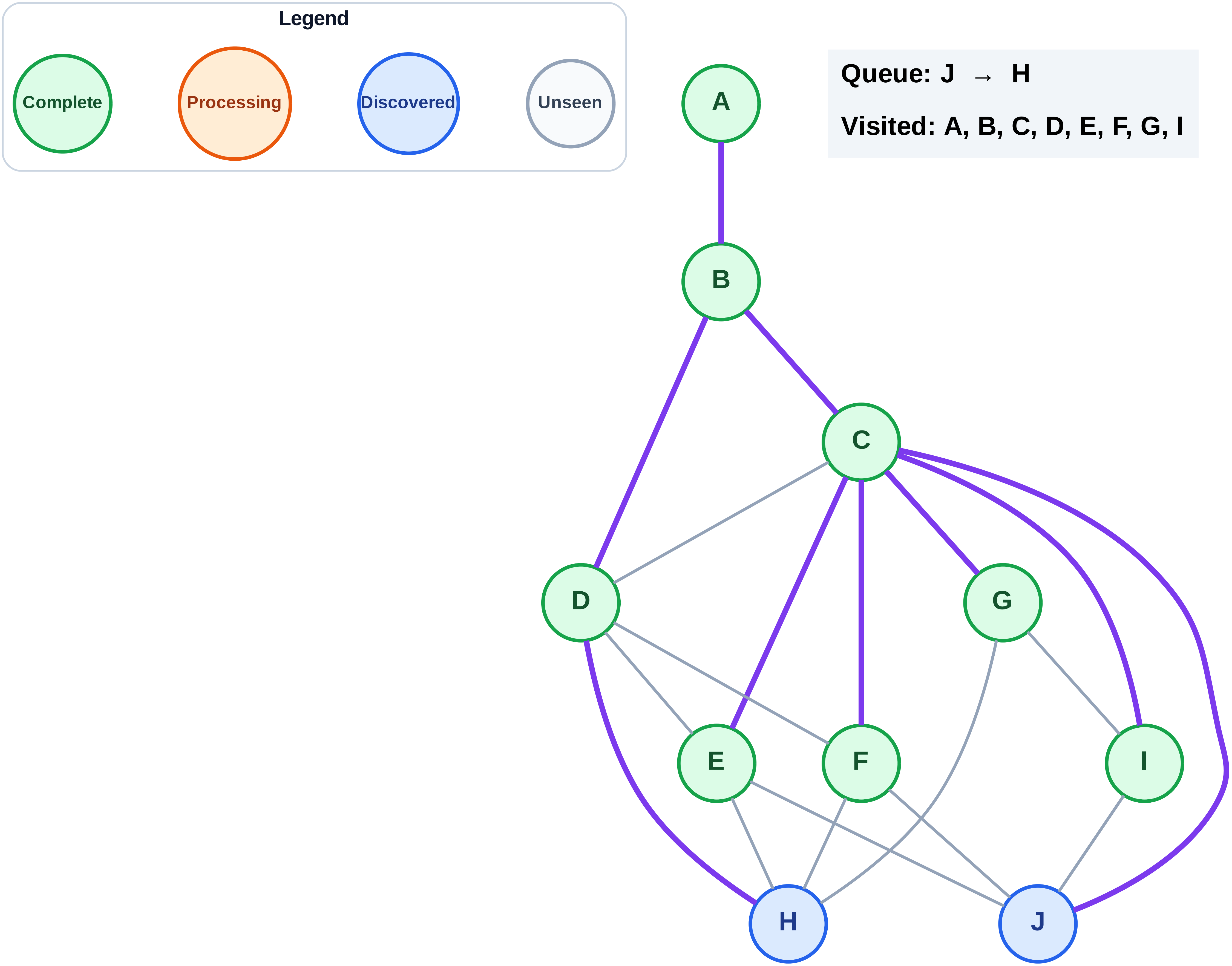
BFS Traversal

Step 16: Process node I



BFS Traversal

Step 17: No new neighbors from I

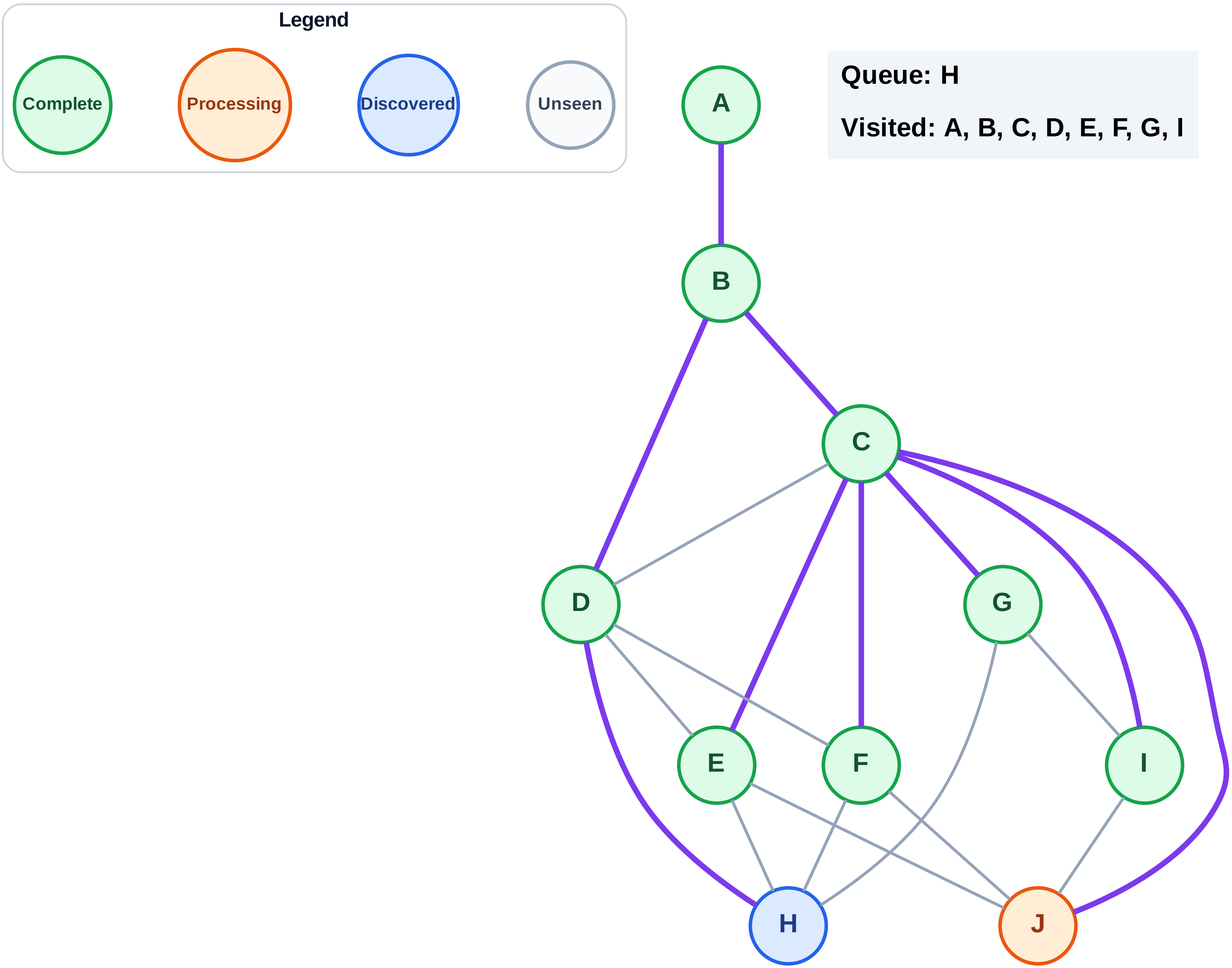


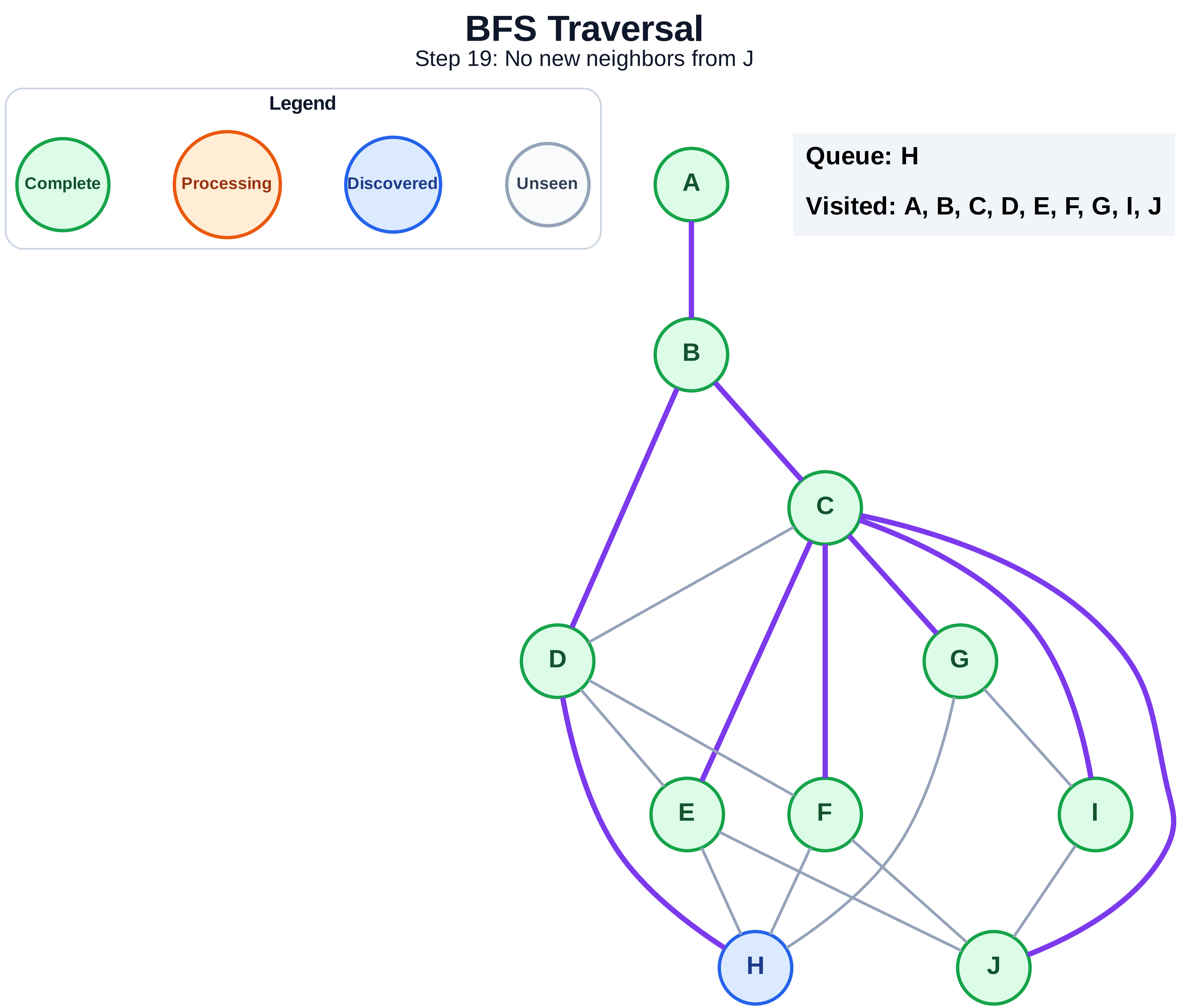
Queue: J → H

Visited: A, B, C, D, E, F, G, I

BFS Traversal

Step 18: Process node J

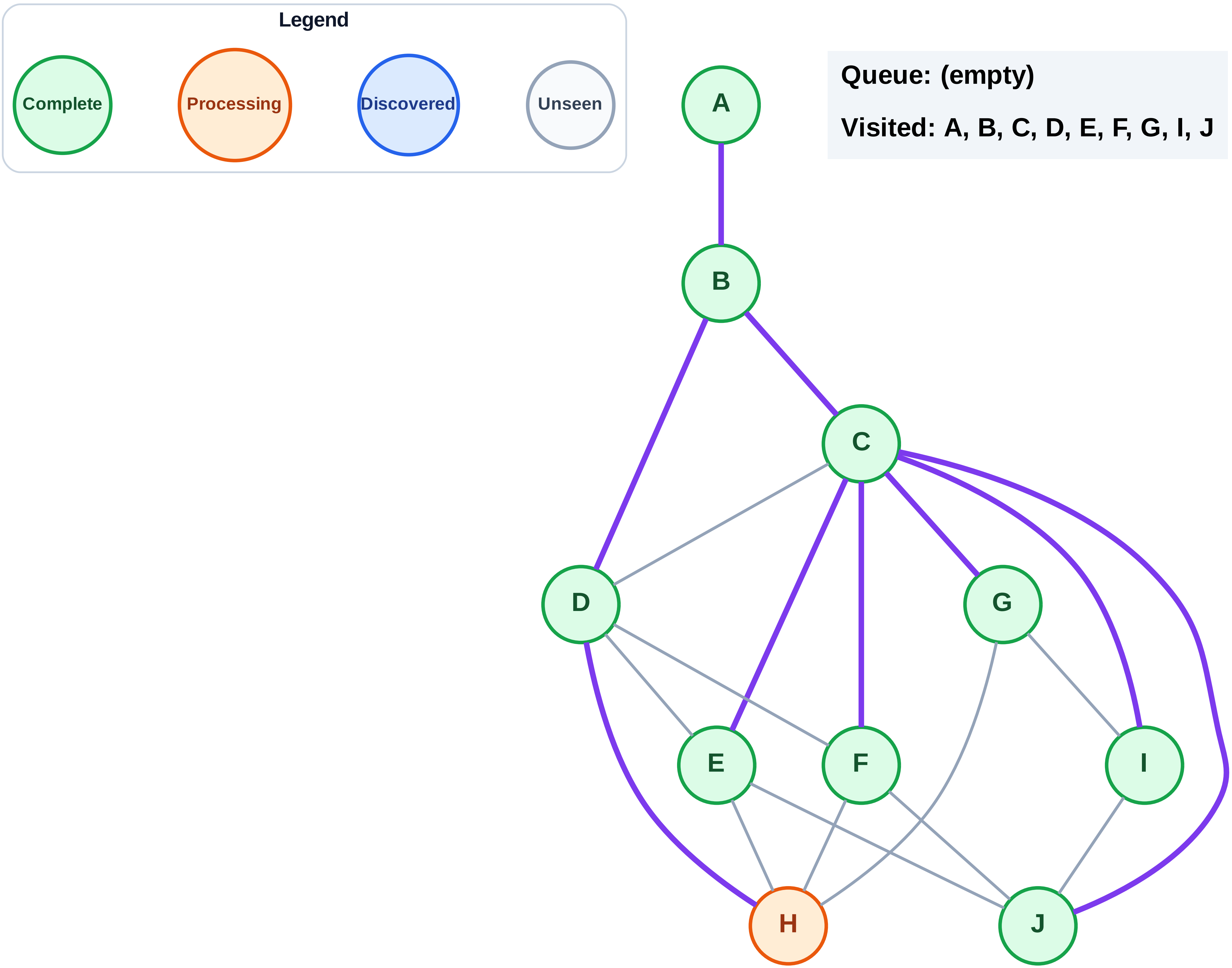




BFS Traversal
Step 19: No new neighbors from J
Legend
Complete
Processing
Discovered
Unseen
Queue: H
Visited: A, B, C, D, E, F, G, I, J
A
B
C
D
G
E
F
I
H
J

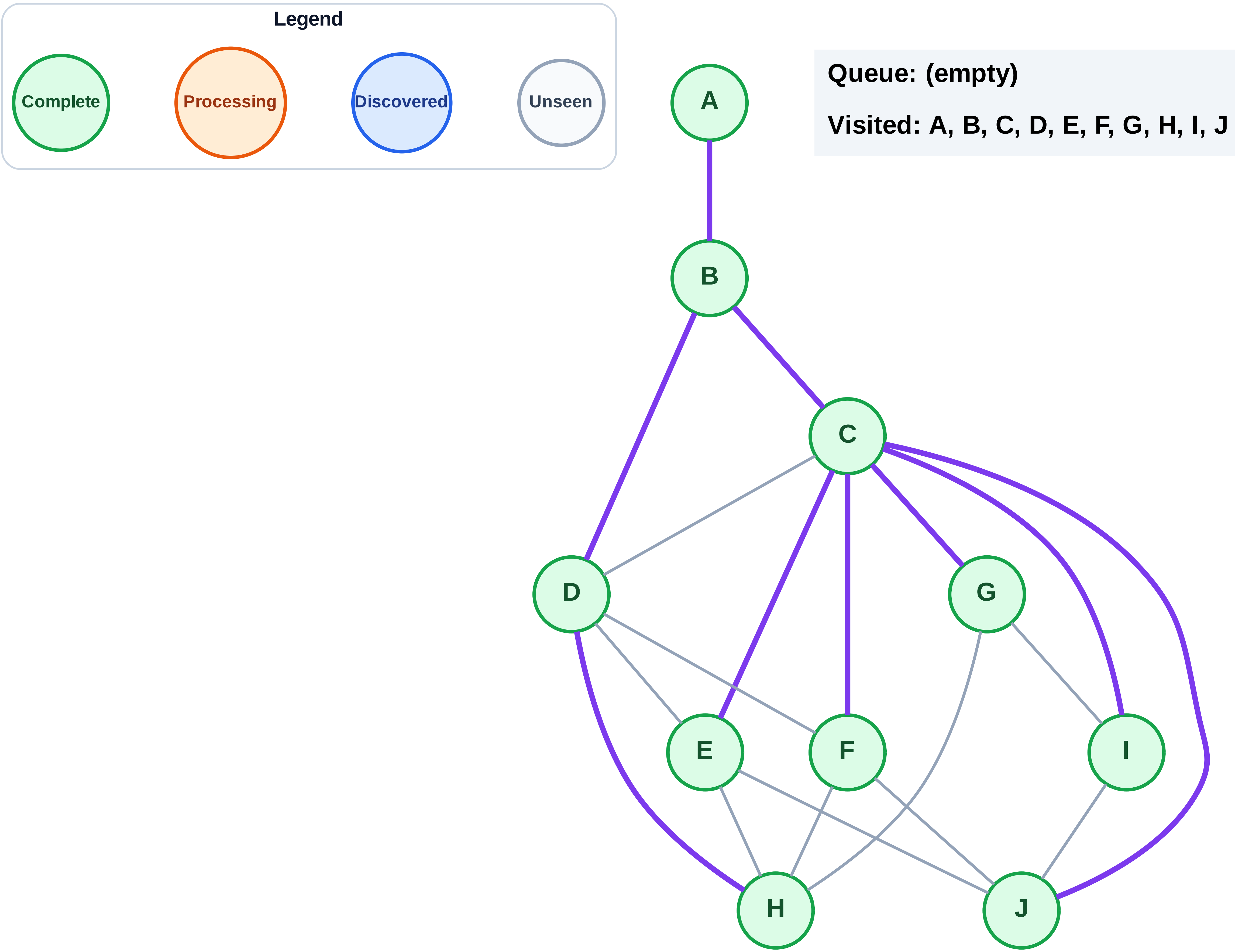
BFS Traversal

Step 20: Process node H



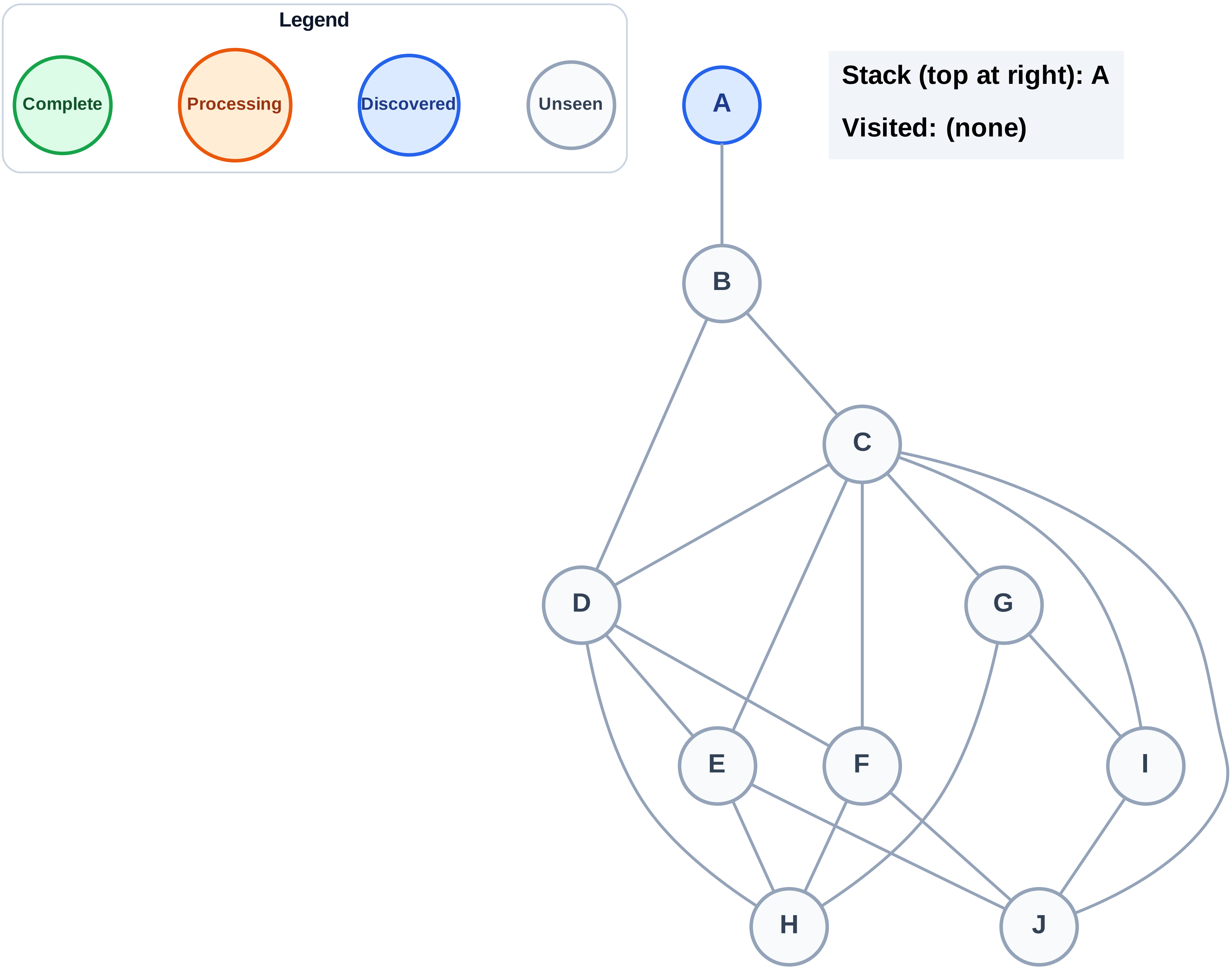
BFS Traversal

Step 21: No new neighbors from H



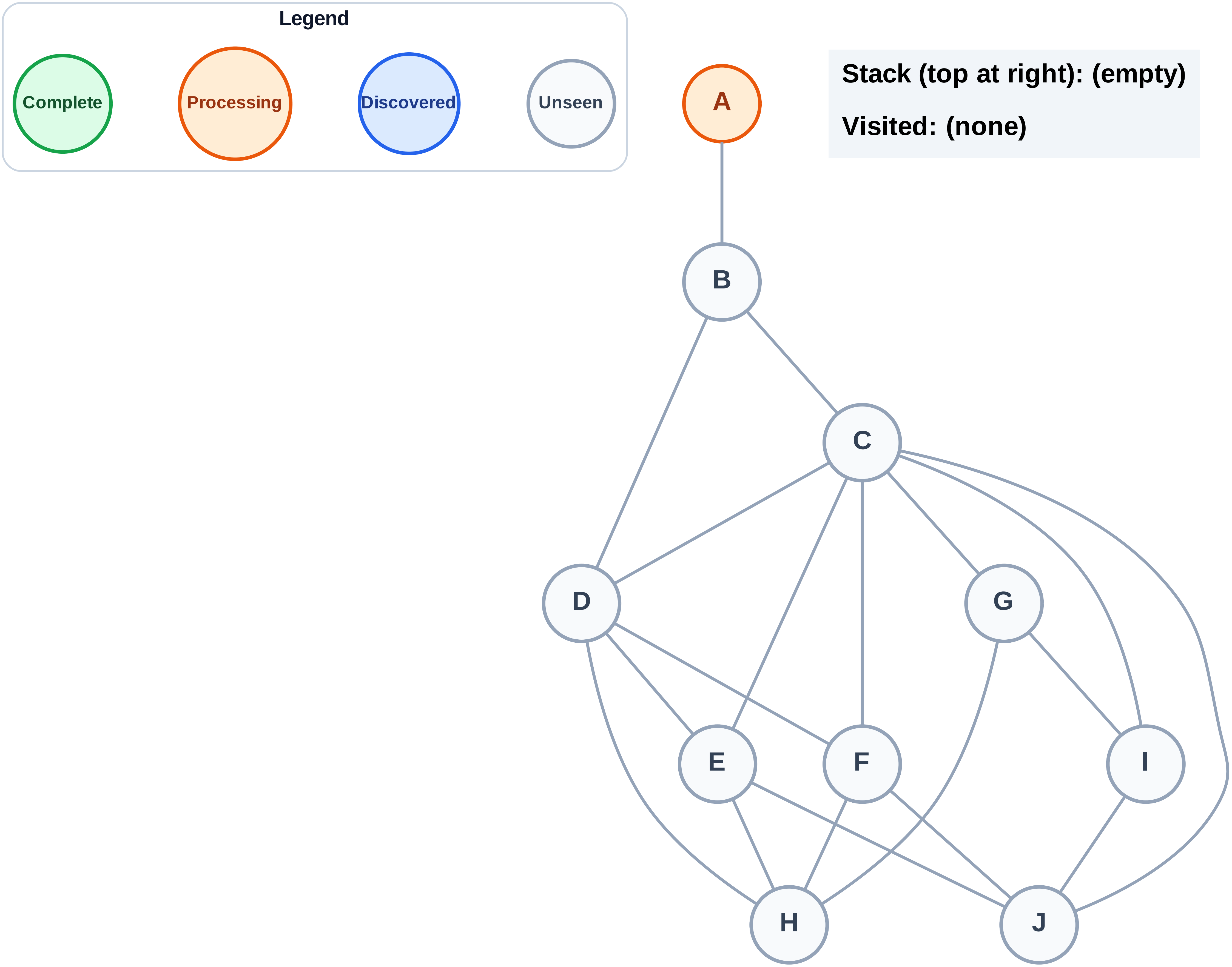
DFS Traversal

Step 1: Start at A



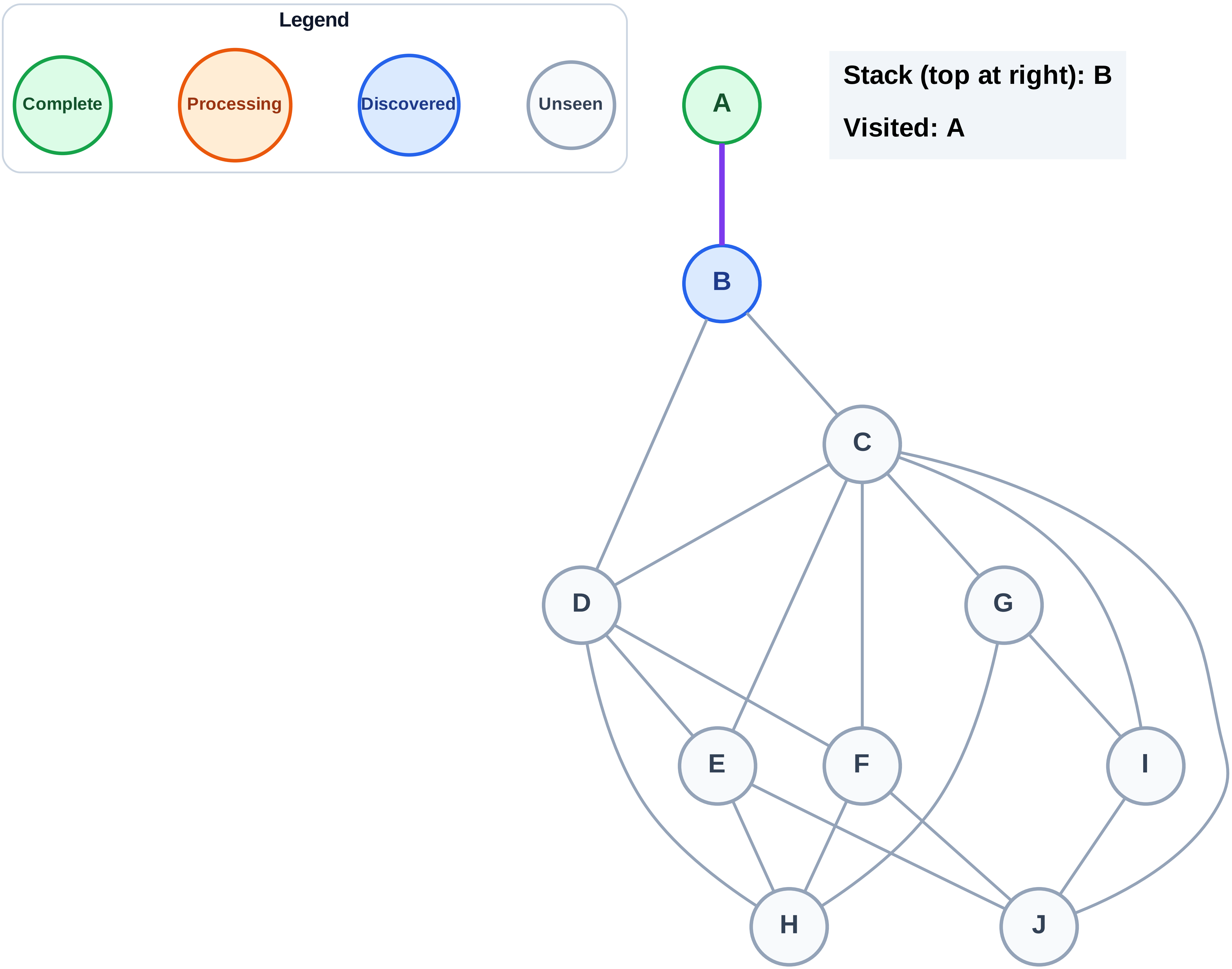
DFS Traversal

Step 2: Process node A



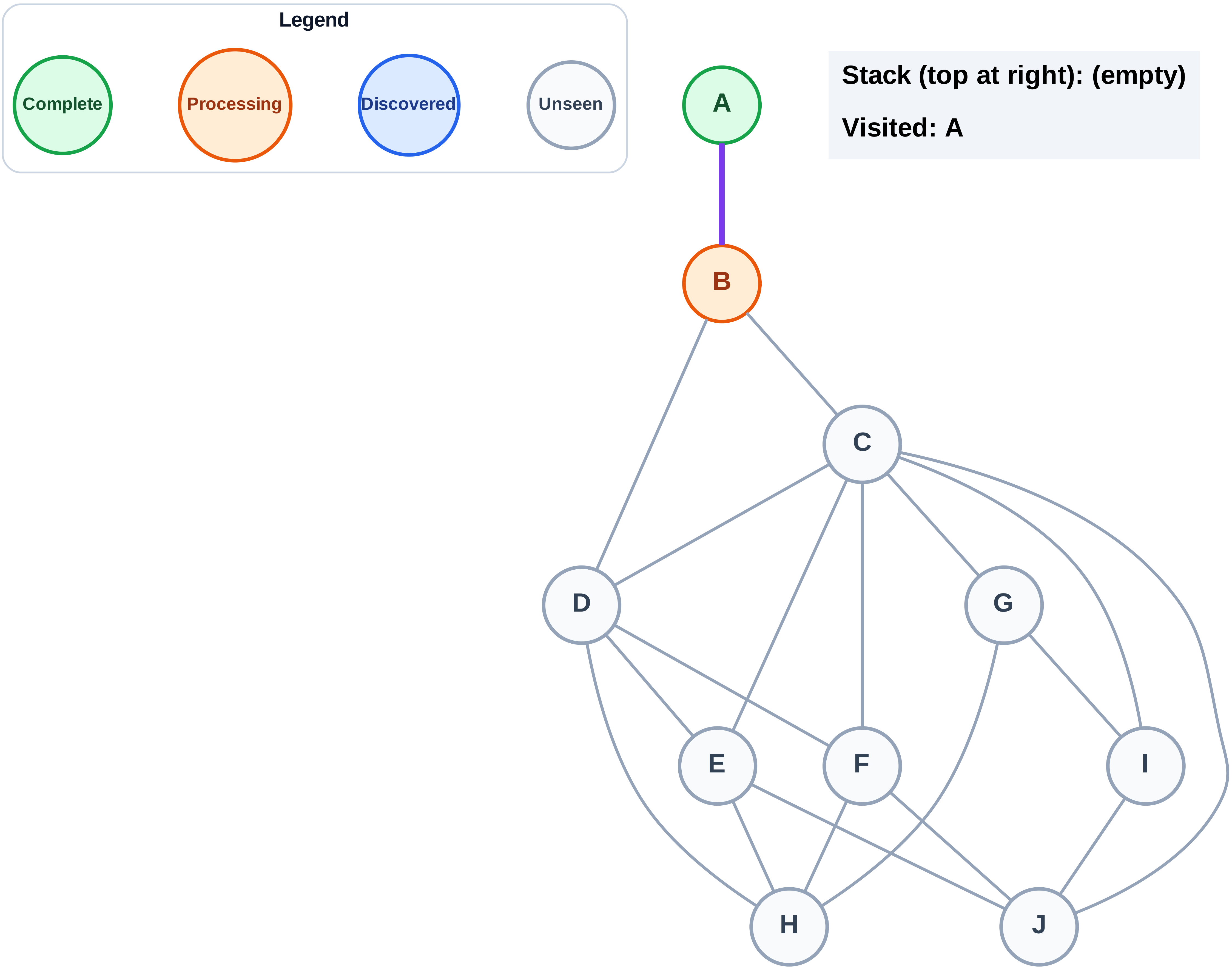
DFS Traversal

Step 3: Discover B from A



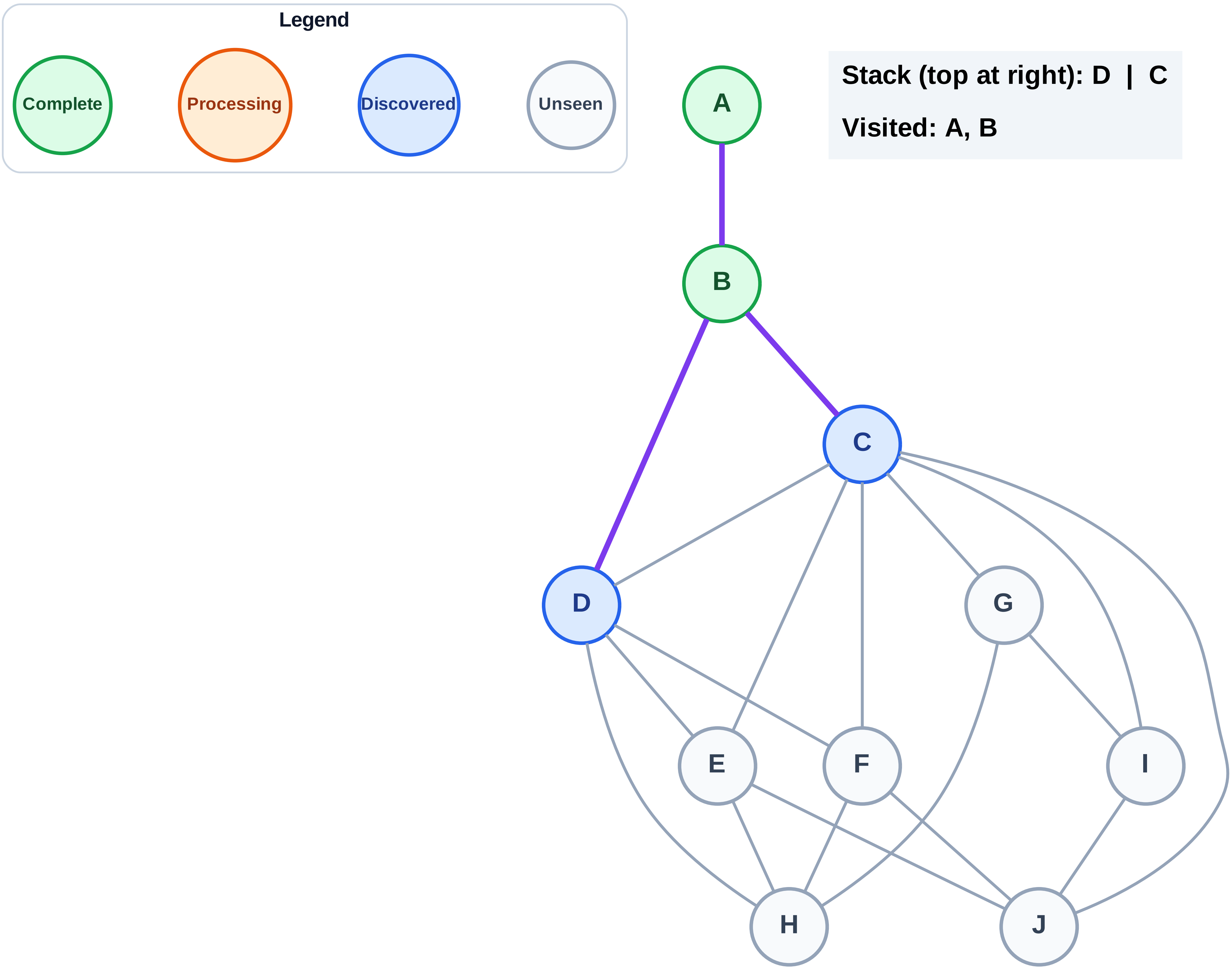
DFS Traversal

Step 4: Process node B



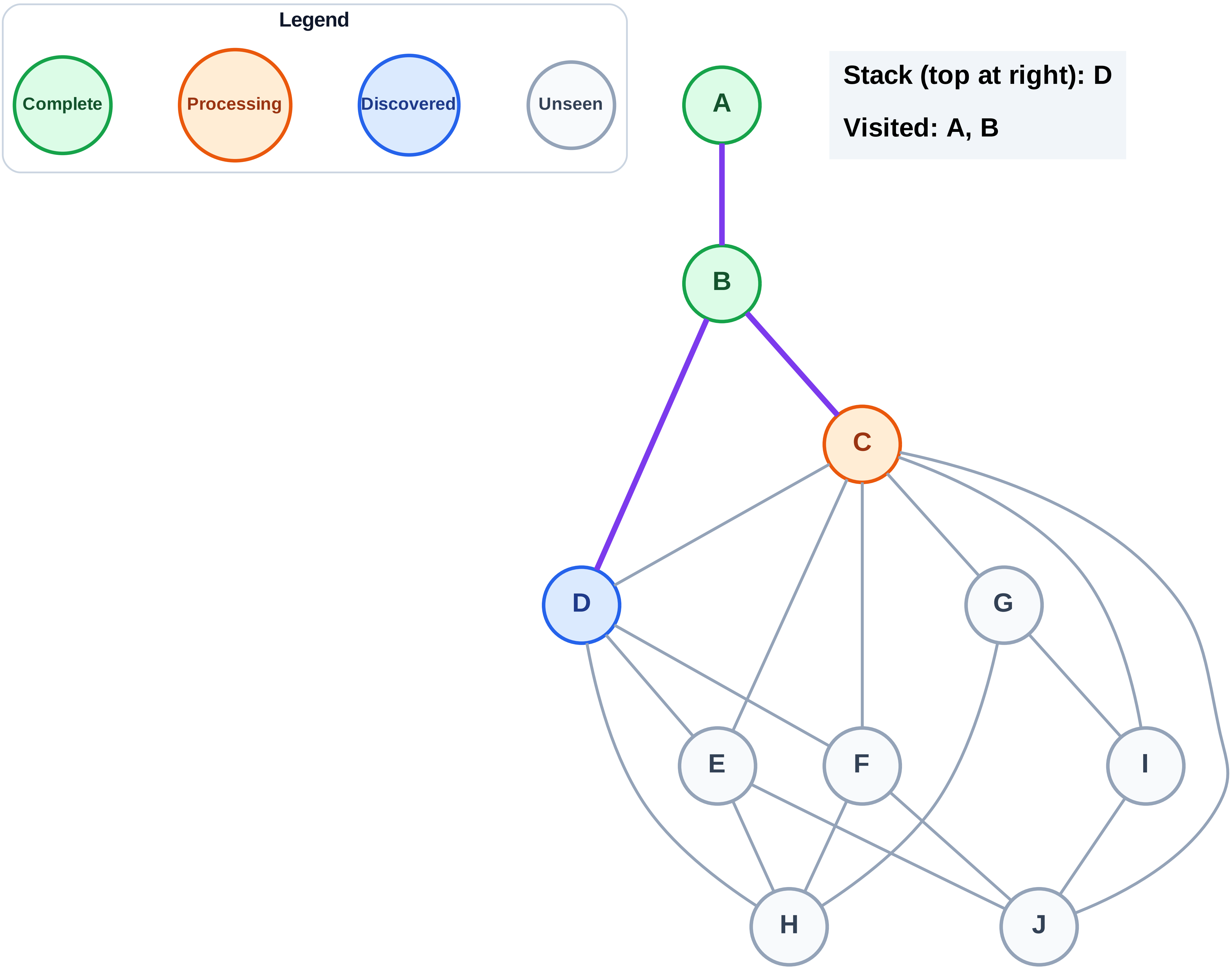
DFS Traversal

Step 5: Discover D, C from B



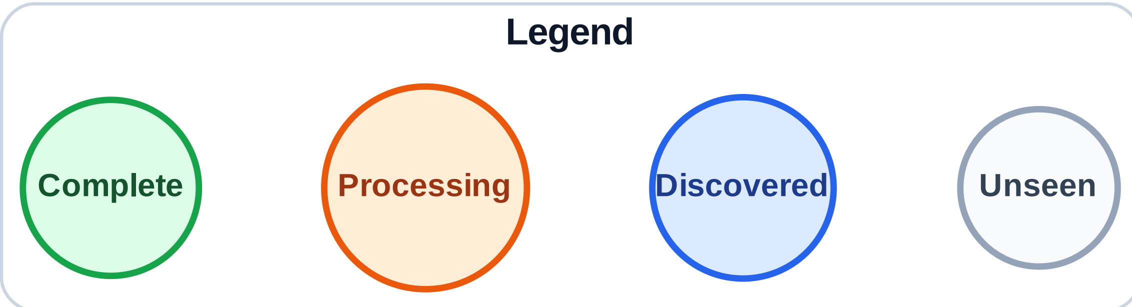
DFS Traversal

Step 6: Process node C



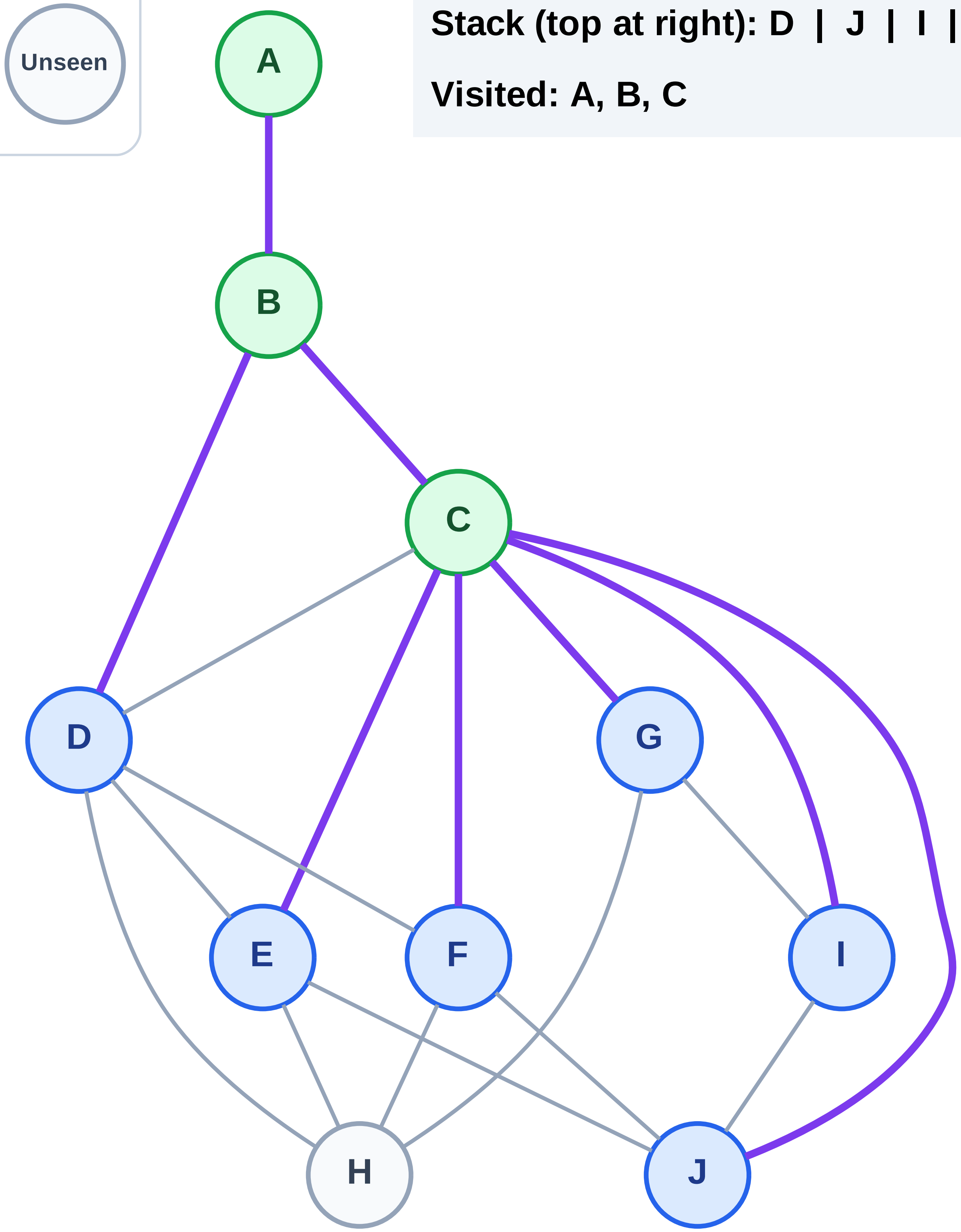
DFS Traversal

Step 7: Discover J, I, G, F, E from C



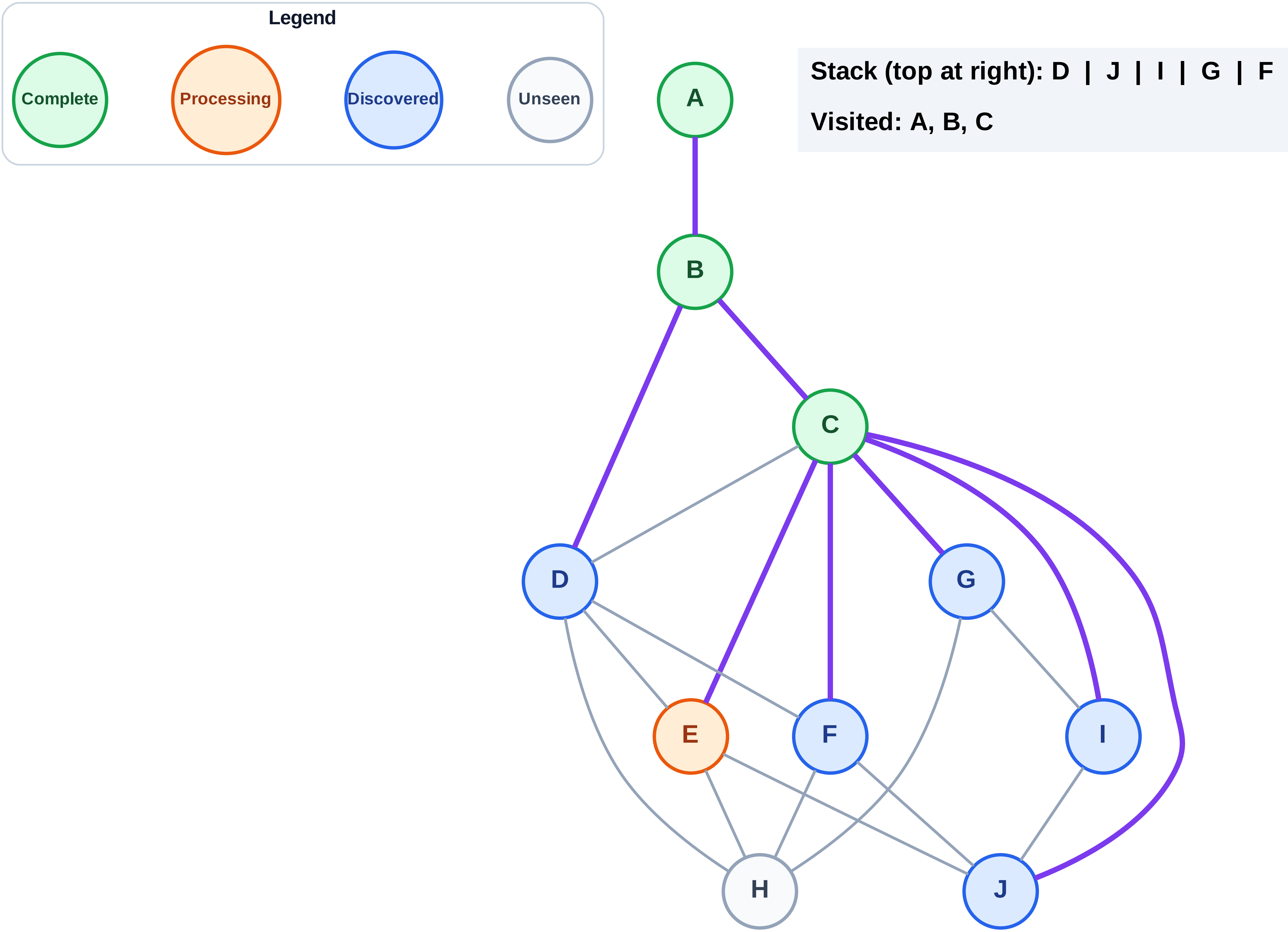
Stack (top at right): D | J | I | G | F | E

Visited: A, B, C



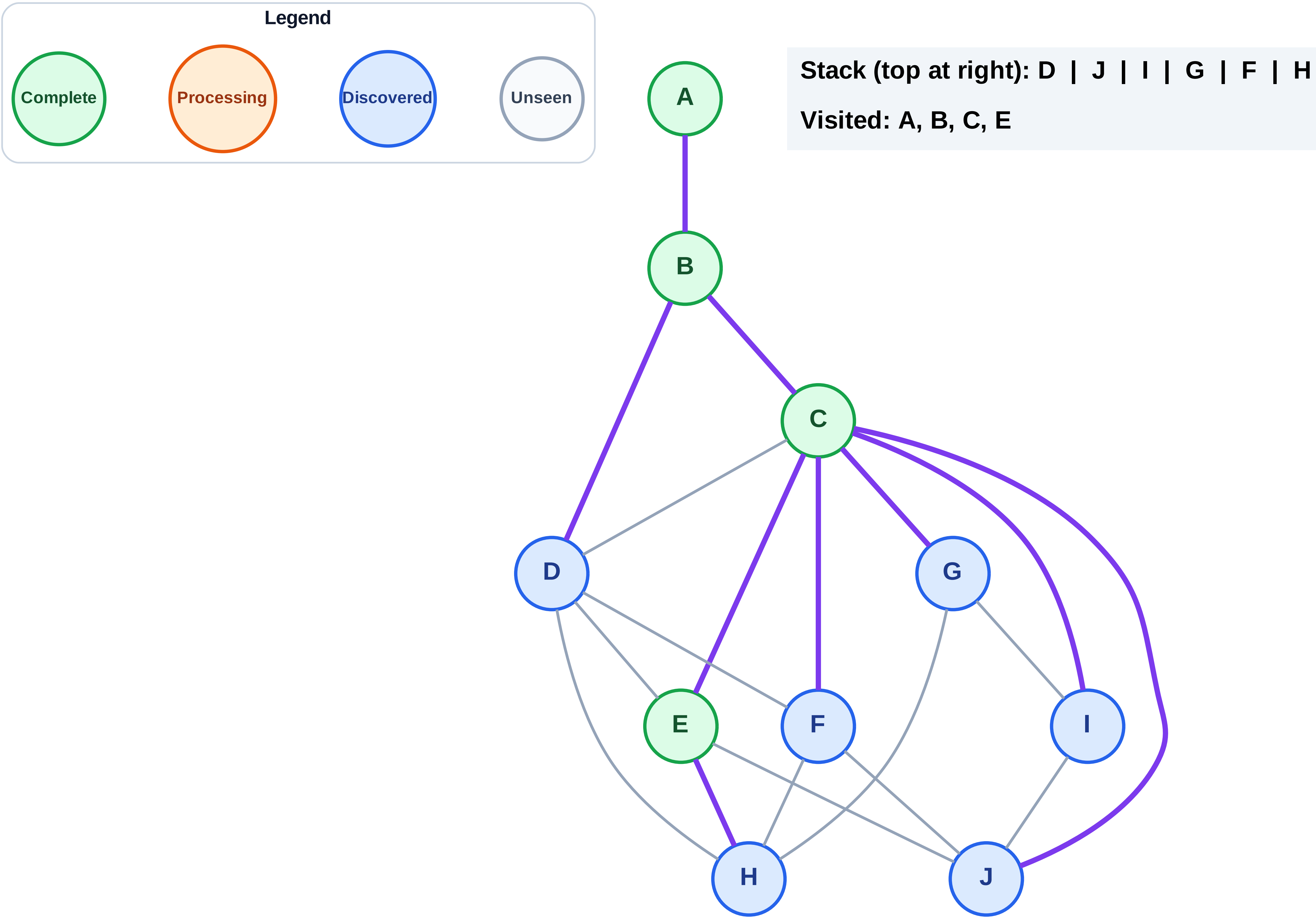
DFS Traversal

Step 8: Process node E



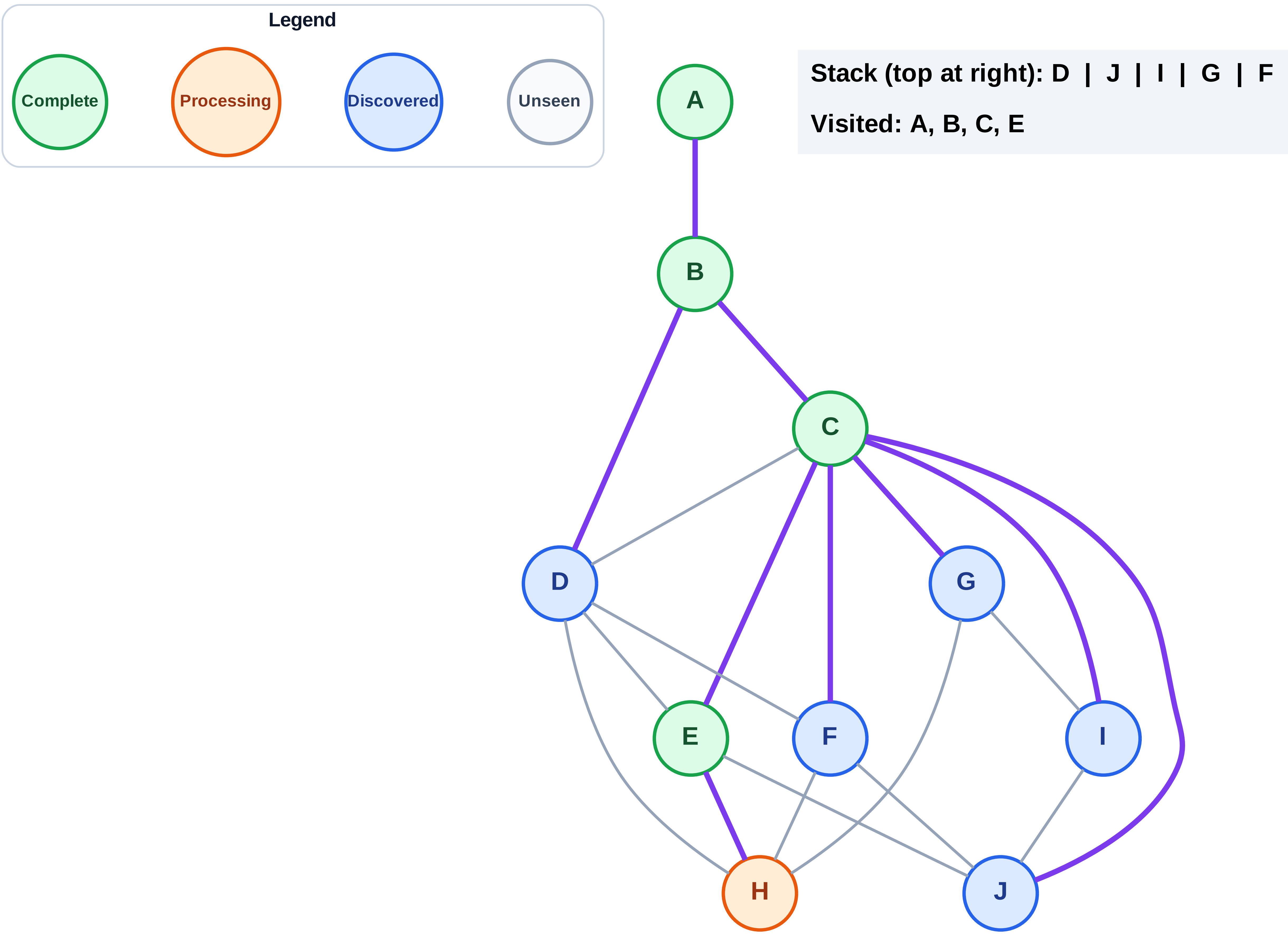
DFS Traversal

Step 9: Discover H from E



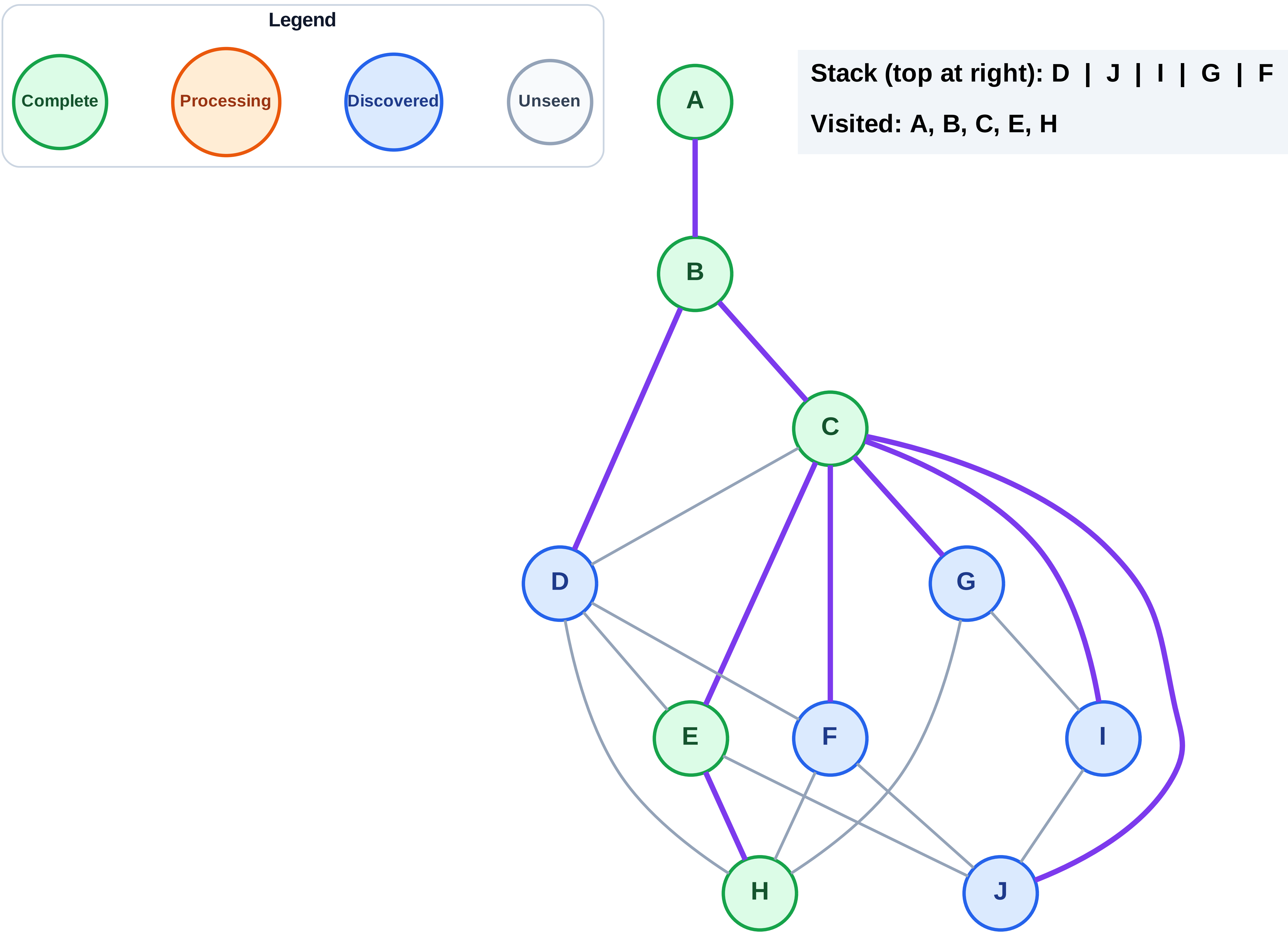
DFS Traversal

Step 10: Process node H



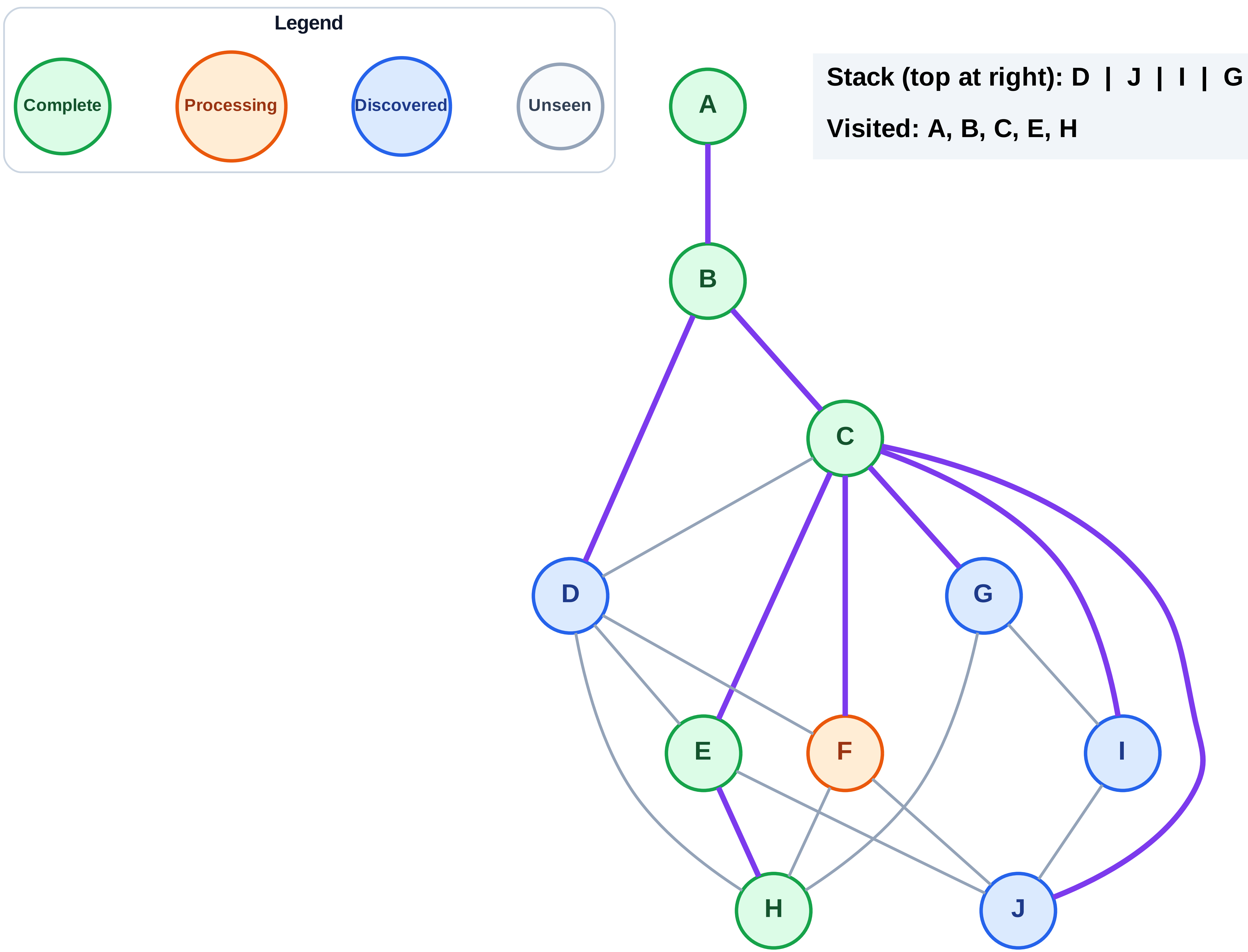
DFS Traversal

Step 11: No new neighbors from H



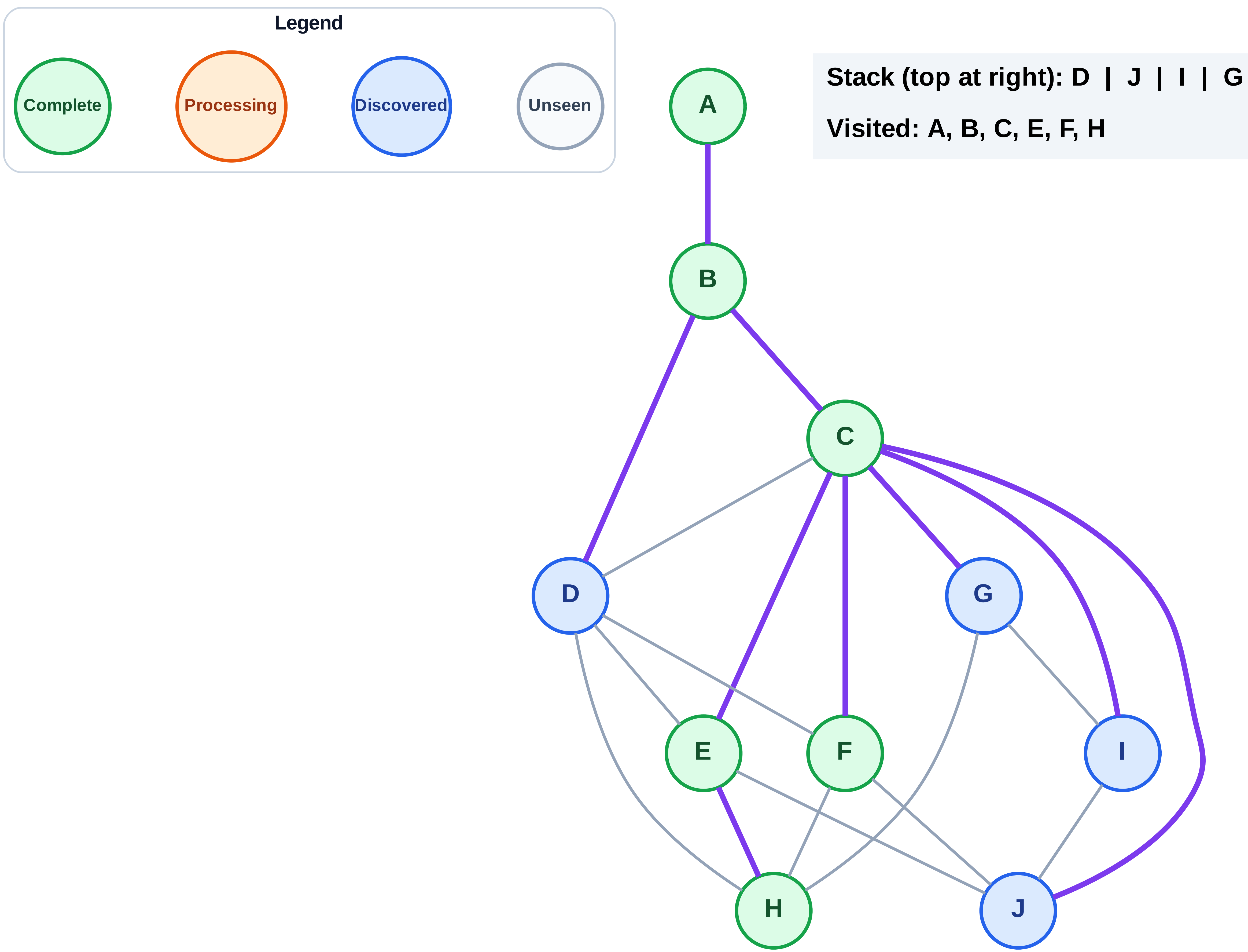
DFS Traversal

Step 12: Process node F



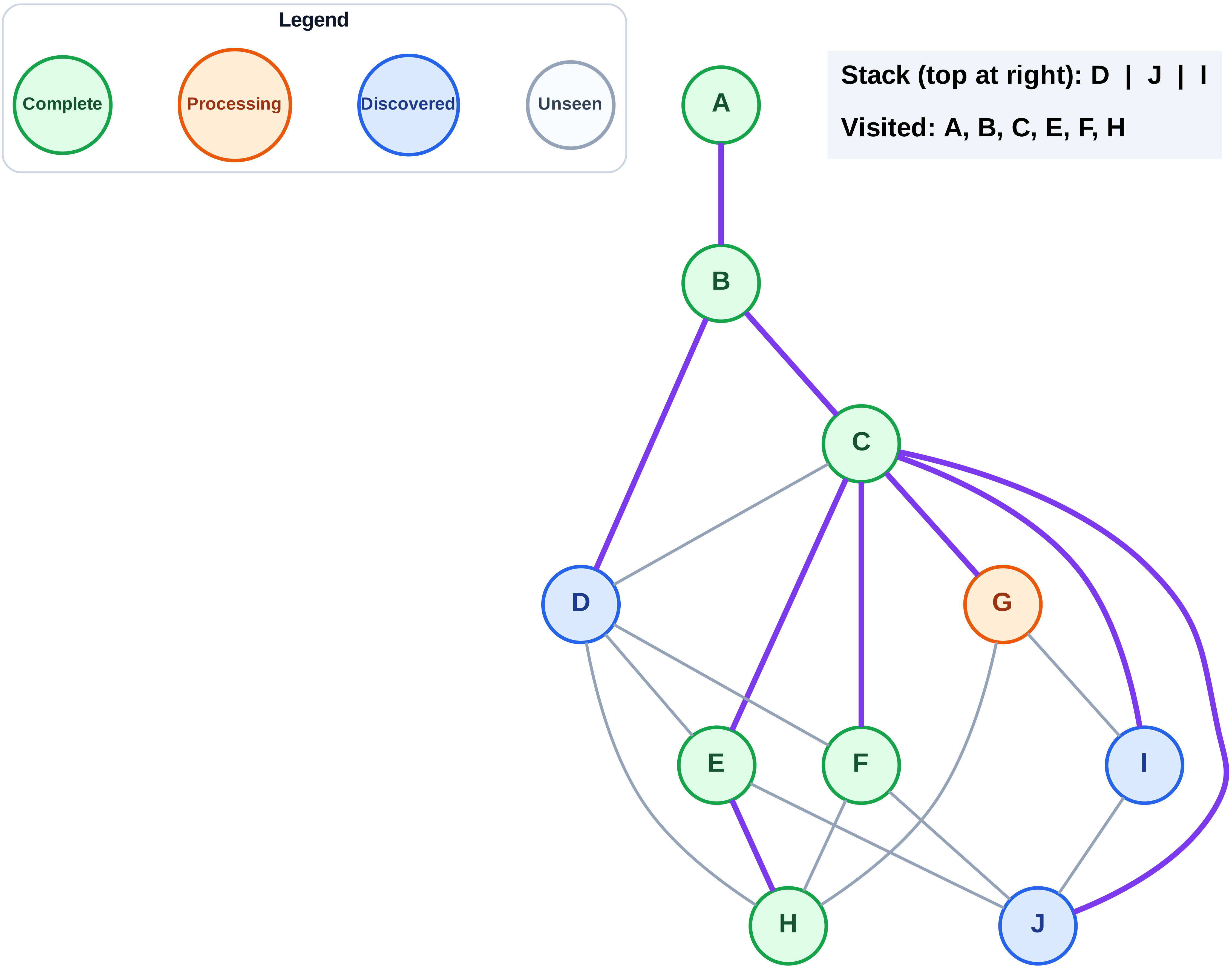
DFS Traversal

Step 13: No new neighbors from F



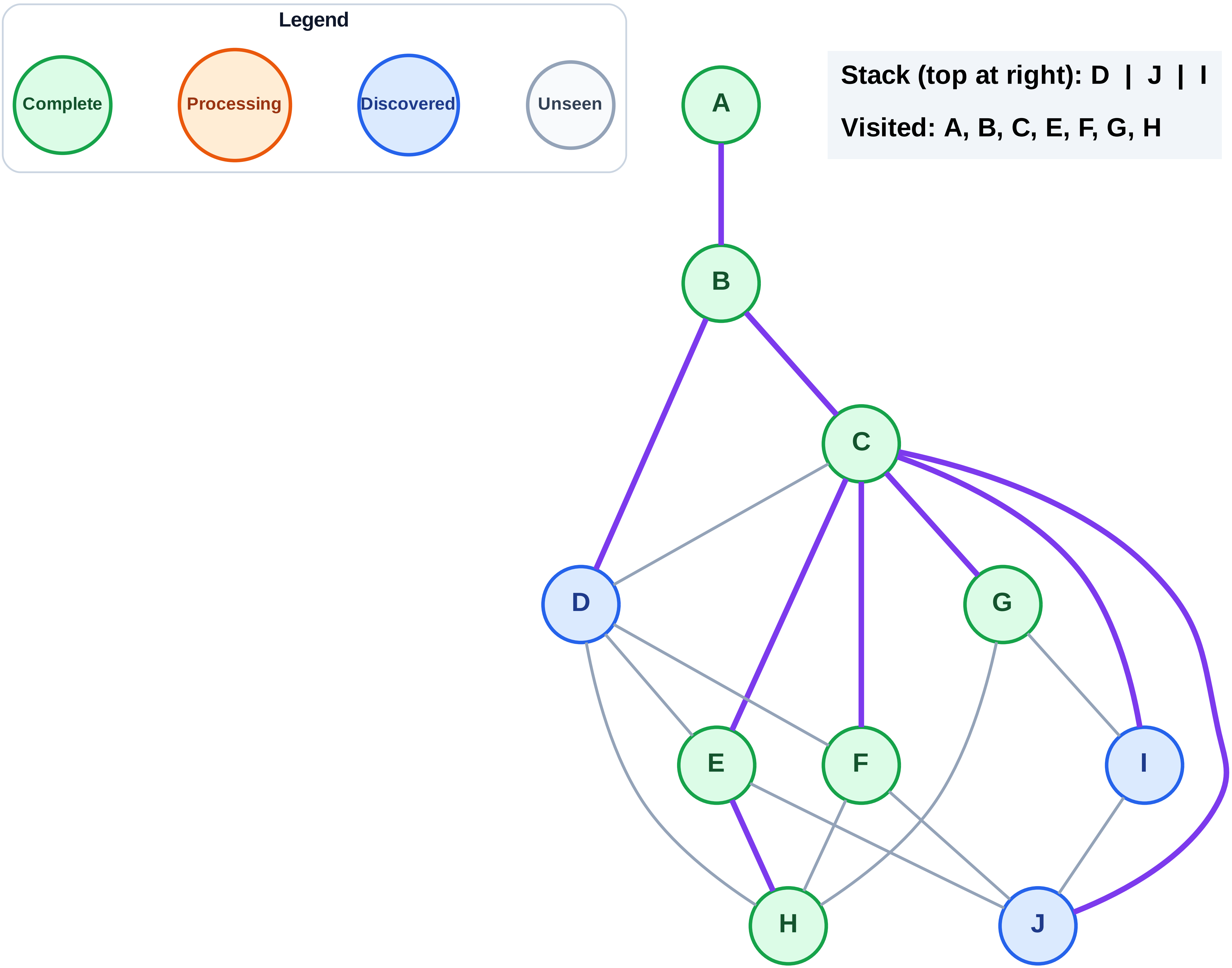
DFS Traversal

Step 14: Process node G



DFS Traversal

Step 15: No new neighbors from G

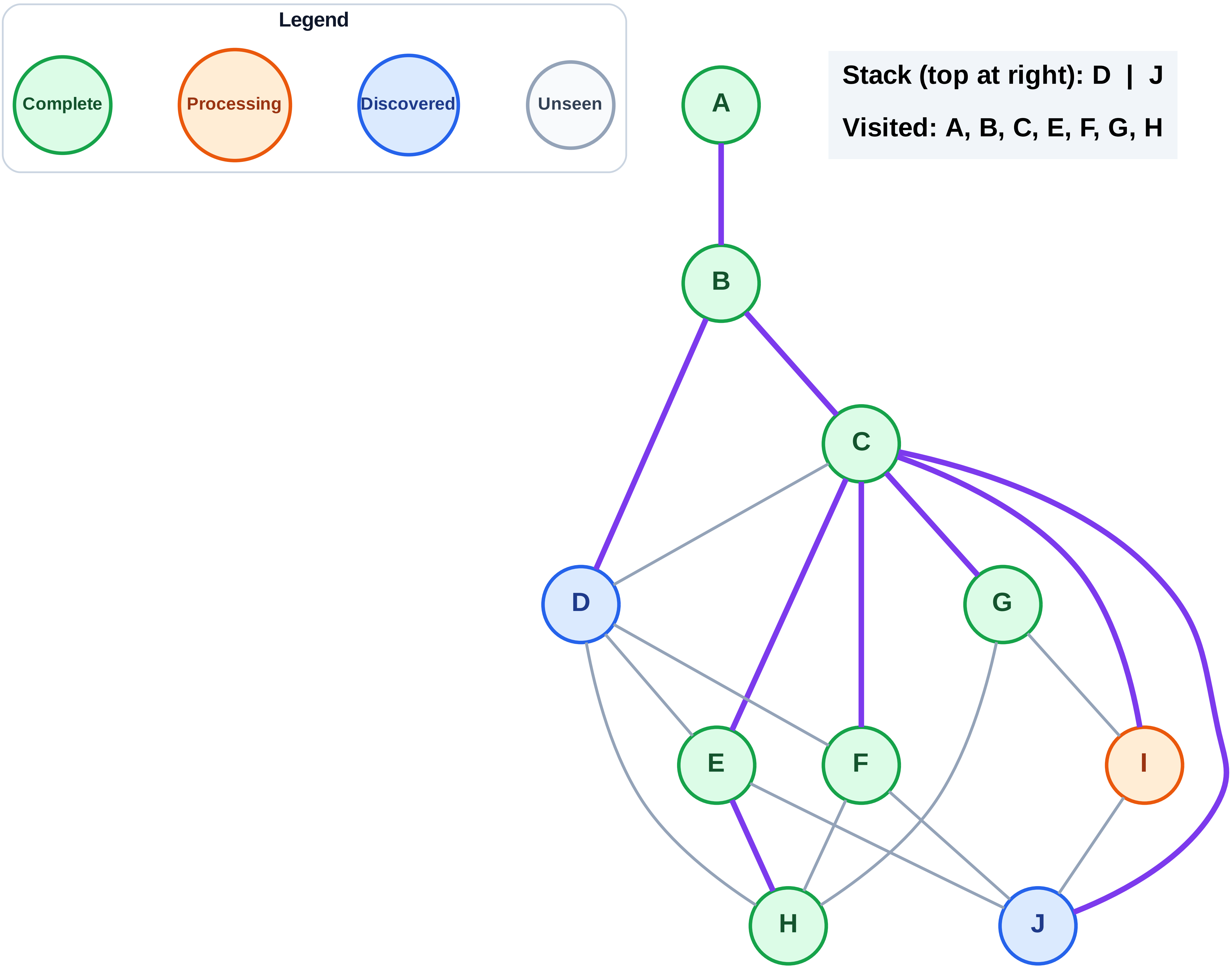


Stack (top at right): D | J | I

Visited: A, B, C, E, F, G, H

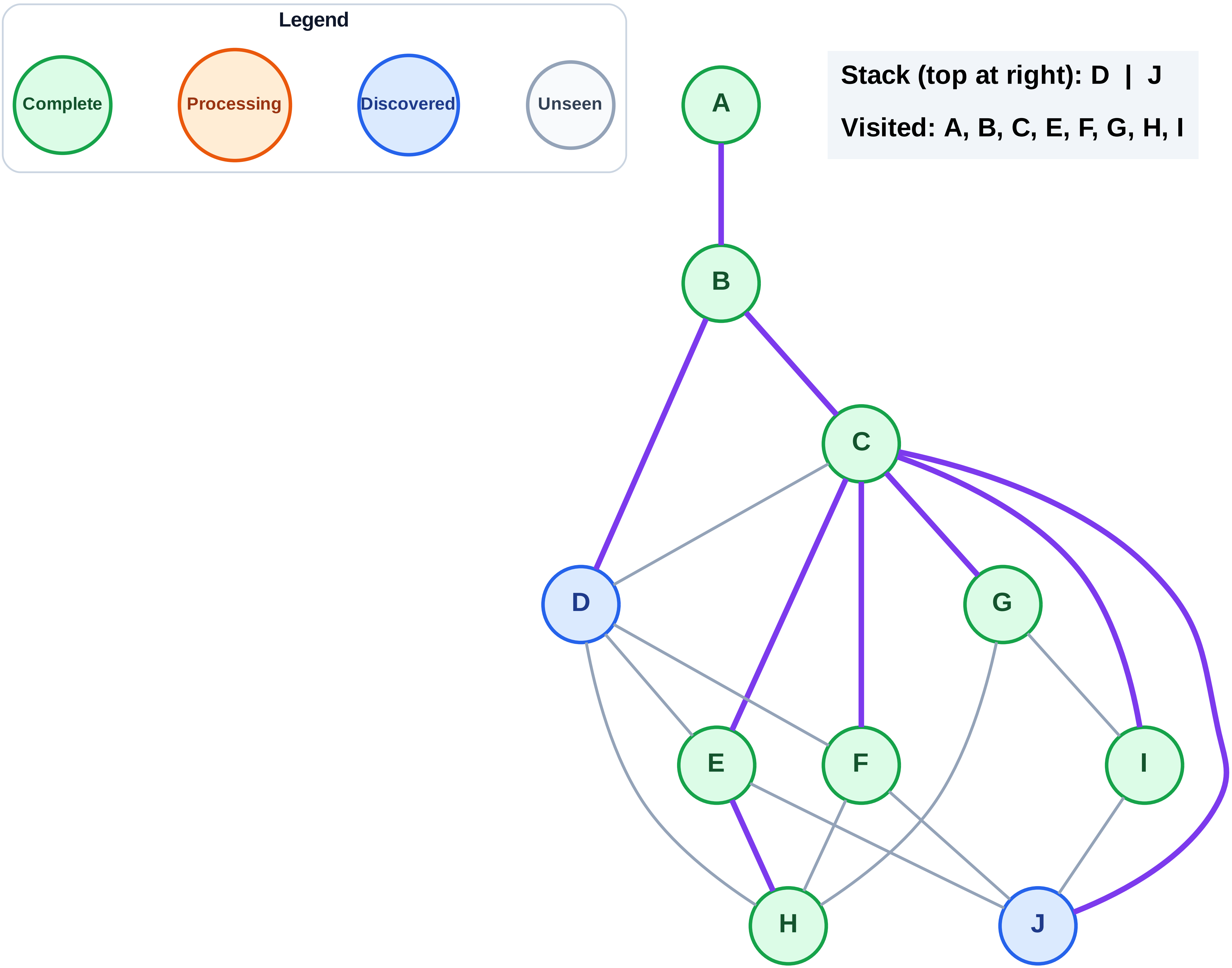
DFS Traversal

Step 16: Process node I



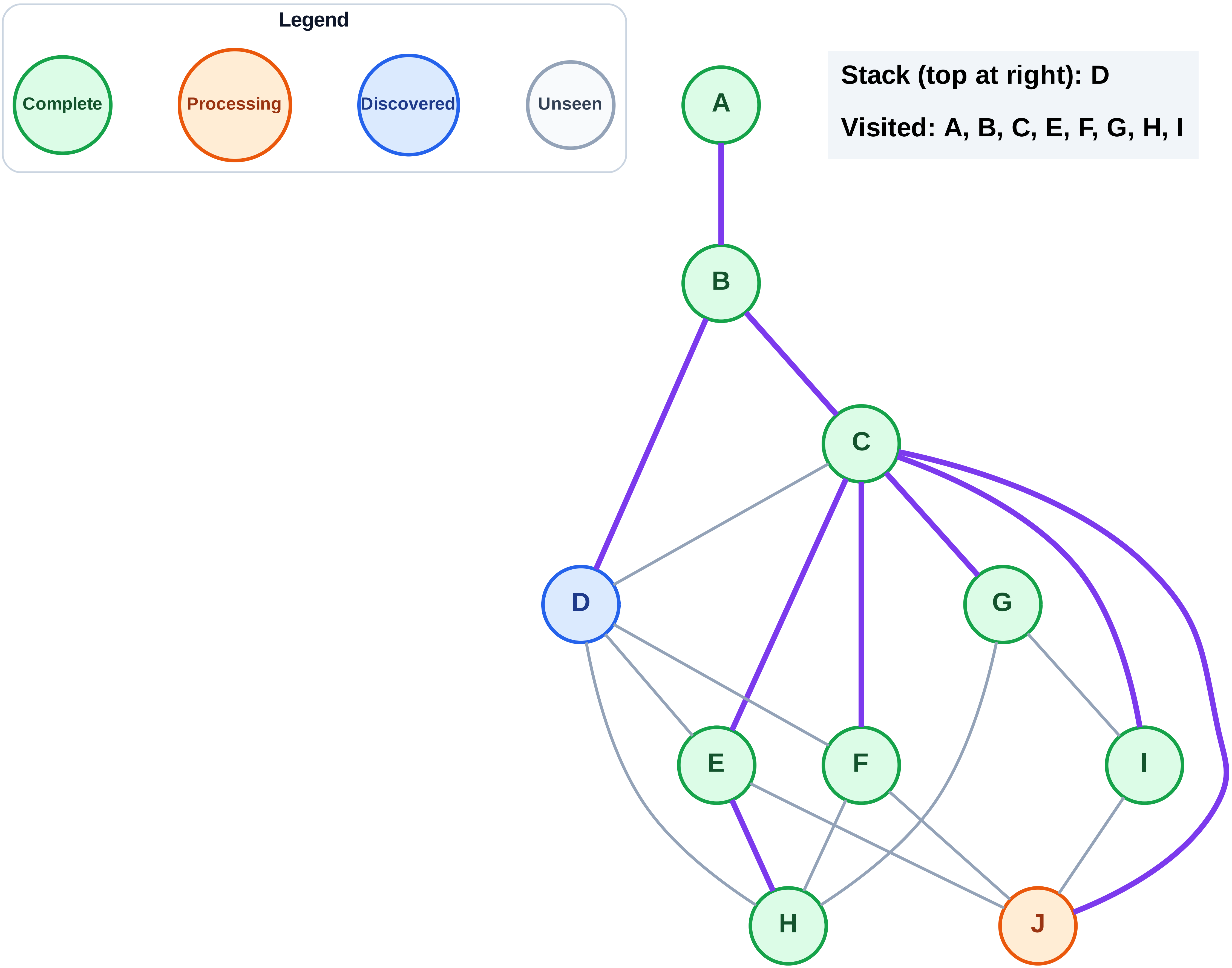
DFS Traversal

Step 17: No new neighbors from I



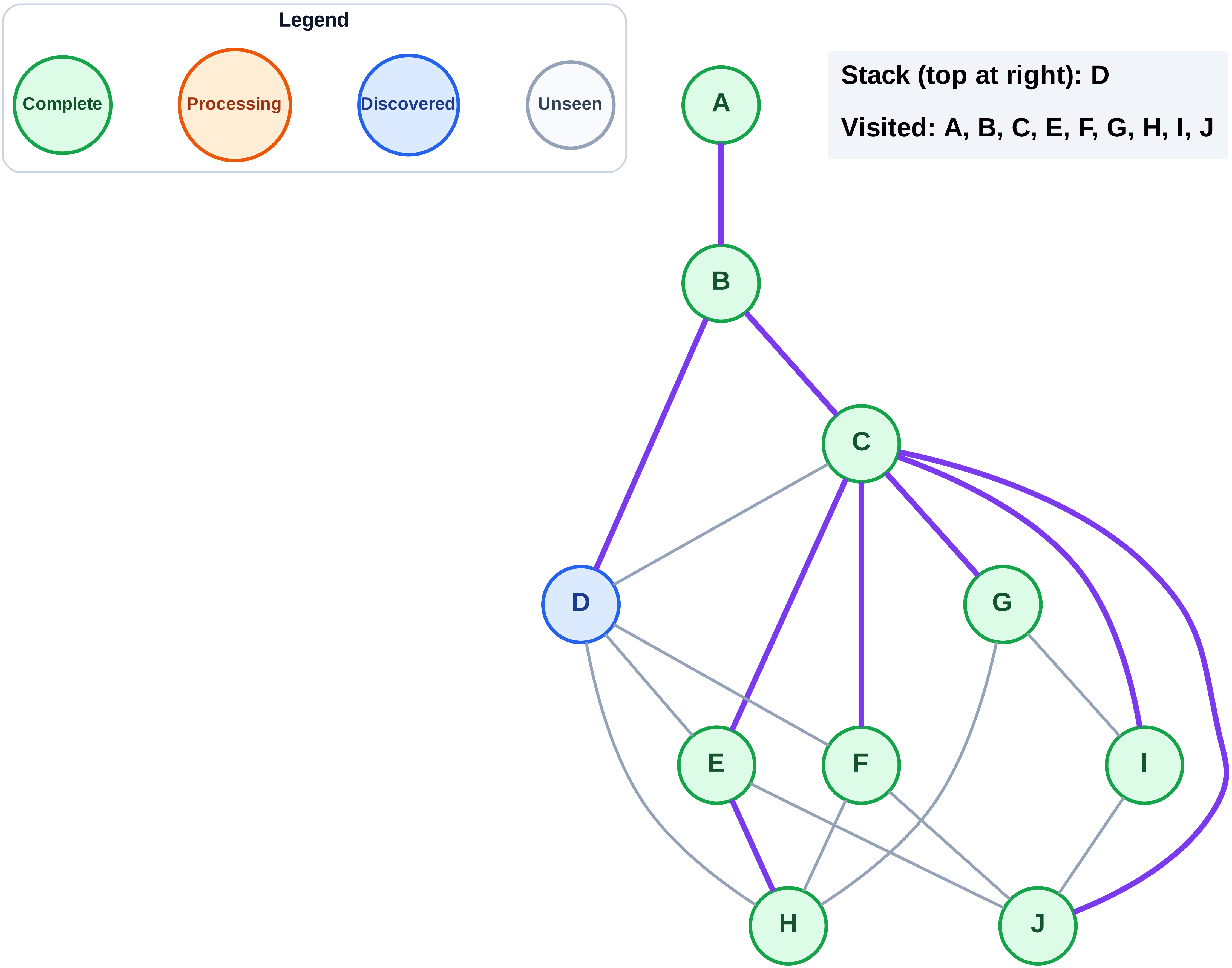
DFS Traversal

Step 18: Process node J



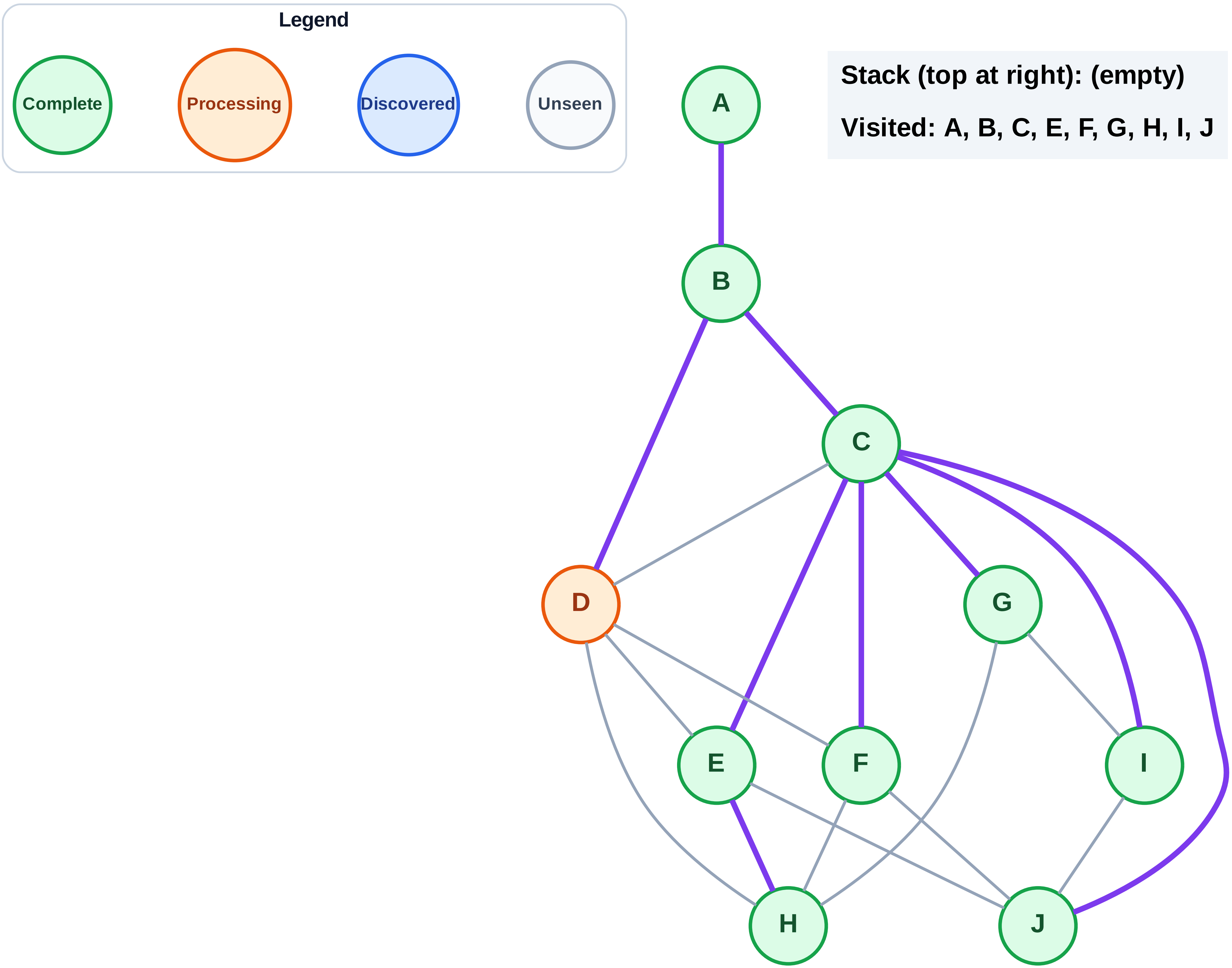
DFS Traversal

Step 19: No new neighbors from J



DFS Traversal

Step 20: Process node D



DFS Traversal

Step 21: No new neighbors from D